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Bridging the Digital Gap in Road Freight Transportation

A Qualitative Study on Why Data-Sharing Platforms Struggle to Gain Traction among Swedish Carriers

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Bridging the Digital Gap in Road Freight Transportation

By

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Master of Science Thesis TRITA-ITM-EX 2026:338
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Att överbrygga den digitala klyftan inom vägtransport av gods

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Abstract

Road Freight Transportation accounts for the largest share of freight movement and CO_2 emissions in the European Union, yet the sector remains characterised by severe operational inefficiencies. Horizontal collaboration offers a potential solution for these inefficiencies but despite the emergence of digital collaboration platforms, practical adoption remains low. This thesis examines the conditions under which data-sharing platforms can support horizontal collaboration in road freight transportation and investigates the factors that sustain the gap between platforms theoretical potential and their uptake in practice.

The study uses three analytical lenses namely the Design-Reality gap framework, Privacy Calculus Theory, and Actor-Network Theory to draw insights into how different factors interact to hinder the transformation. An abductive, qualitative research design was used, combining a systematic literature review with semi-structured interviews.

The findings reveal that the sector largely remains computerised rather than digitalised and most of the transport coordination is happening via phone and email. In addition, data sensitivity emerges as a relational rather than a categorical property which means that the carriers' willingness to share data is determined not by the data itself, but by the governance context in which it is shared. The tacit knowledge held by experienced dispatchers further limits the replicability of existing platforms. Lastly, the governance structure of the Swedish Lastbilscentral is analysed and identified as a potential resolution for effective platform adoption.

These findings extend the Privacy Calculus Theory by redefining data sensitivity as context dependent and contribute to the Actor-Network Theory by positioning governance structure as a prerequisite for network formation. The design-reality gap is reframed as multi-dimensional and interconnected rather than reducible to a single cause.

The results suggest that platforms, carriers and policymakers should prioritise governance architecture over privacy features to accelerate platform adoption.

Keywords: road freight transportation, horizontal collaboration, data sharing, design-reality gap, privacy calculus, Actor-Network Theory, governance, digitalisation

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Sammanfattning

Vägtransporter står för den största andelen av godstransporterna och koldioxidutsläppen inom Europeiska unionen, men sektorn präglas fortfarande av driftmässig ineffektivitet. Horisontellt samarbete erbjuder en möjlig lösning för att öka effektiviteten, men trots framväxten av digitala samarbetsplattformar är användningen av dessa fortfarande låg. Denna avhandling undersöker under vilka förutsättningar plattformar för datadelning kan stödja horisontellt samarbete inom vägtransportsektorn och kartlägger de faktorer som upprätthåller klyftan mellan plattformarnas teoretiska potential och deras praktiska användning.

Studien använder tre analytiska perspektiv, nämligen Design-Reality Gap, Privacy Calculus Theory och Actor-Network Theory, för att få insikter i hur olika faktorer fungerar som barriärer för omvandlingen. En abduktiv, kvalitativ forskningsdesign användes, där en systematisk litteraturöversikt kombinerades med semistrukturerade intervjuer.

Resultaten visar att sektorn i stor utsträckning är datoriserad snarare än digitaliserad och att merparten av transportplaneringen sker via telefon och e-post. Dessutom framstår datakänslighet som en relationell snarare än en kategorisk egenskap, vilket innebär att transportörernas vilja att dela data inte avgörs av själva datan, utan av det styrningssammanhang inom vilket den delas. Den tysta kunskapen som erfarna trafikledare besitter begränsar ytterligare möjligheten att replikera befintliga plattformar. Slutligen analyseras styrningsstrukturen hos svenska lastbilcentraler och identifieras som en potentiell lösning för effektivt plattformsinförande.

Dessa resultat utvidgar Privacy Calculus Theory genom att omdefiniera datakänslighet som kontextberoende och bidrar till Actor-Network Theory genom att positionera styrningsstrukturen som en förutsättning för nätverksbildning. Design-Reality Gap omformuleras som flerdimensionellt och sammanlänkat snarare än reducerbart till en enda orsak.

Resultaten tyder på att plattformar, transportörer och beslutsfattare bör prioritera styrningsarkitekturen framför integritetsfunktioner för att påskynda plattformsinförandet.

Nyckelord: vägfrakt, transport, horisontellt samarbete, datadelning, klyftan mellan design och verklighet, integritetskalkyl, aktör-nätverksteori, styrning, digitalisering

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Abbreviations

ANT	Actor-Network Theory
DP	Digital Platform
EU	European Union
FTL	Full Truckload
HC	Horizontal Collaboration
HCT	Horizontal Collaborative Transport
LBC	Lastbilscentral
LTL	Less Than Truckload
RFT	Road Freight Transport
TMS	Transport Management System
VC	Vertical Collaboration

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1 Introduction

Road freight transport (RFT) plays a fundamental role in our modern society as it enables the movement of goods across supply chains stretching over multiple continents until an end-product reaches the customer. In recent years as global trade continued to expand by another $2.2 trillion in 2025, the demand for efficient and reliable supply chains and hence logistics has continuously grown (United Nations Trade and Development, 2025). At the same time, the transport sector needs to comply with an increasing number of regulations that demand a reduction in its environmental impact. Especially in Europe the RFT accounts for a majority of the transported volume whilst also having the highest environmental impact compared to other economically viable transport modes like sea or rail transport (Skrucany *et al.*, 2018; Samuelson, 2021).

Despite the technological improvements and alternative eco-friendly transport modes, the current share of diesel heavy duty vehicles is still 94% and zero emission trucks only account for a small single digit percentage of the registered trucks in the European Union (EU) (Grzelakowski, 2025; Ragon, 2025; Nogueira and Lamarque, 2026). The rate at which new zero emission trucks are registered varies drastically between the different countries in the EU where Norway, Sweden, Netherlands and Denmark register the most number of new electric heavy trucks compared to the general registrations in the EU (Nogueira and Lamarque, 2026). This slow adoption rate of more sustainable modes of transportation generates a demand for other ways to improve the efficiency of the RFT.

One of the biggest inefficiencies in the RFT lies in the low utilisation of the vehicles. A fifth (21,6%) of all road freight driven kilometres in the EU was driven by completely empty vehicles (Eurostat, 2024). These so-called empty runs are a major driver for inefficiencies because they do not only not generate any value by transporting goods but instead consume additional fuel, generate emissions and contribute further to congested roads. Empty runs have been an issue in the industry for a long time but were widely considered to be unavoidable. The recent demand for more efficient solutions has however brought the discussions about a potential reduction of empty runs back and it is now more relevant than ever.

Increased collaboration between carriers that would allow them to better utilise their assets has shown to be the most mentioned approach in the literature. However, despite documented efficiency gains this collaboration has remained limited in scope and scale. The persistence of this gap between what platforms theoretically offer and what carriers adopt suggests that the barriers are embedded in structural and institutional behaviours. This study examines this design-reality gap through interviews with Swedish road

freight carriers and industry experts, with the aim of identifying existing structures that hinder this implementation and what needs to happen for these systems to gain relevance.

The project is carried out in a collaboration with VTI, Statens väg- och transportforskningsinstitut, an independent research institute in the transport sector (https://www.vti.se/).

1.1 Background

As mentioned before the RFT plays an important role in the European economy but there are structural inefficiencies that need to be addressed which will be shortly introduced in the following.

Currently the whole logistics sector generates 7% of the GDP of the European Union, while the freight transport itself contributes to 2.5% of the total GDP. Considering that the freight transport only plays a supporting role in the actual value creation, the impact is relatively high (Fazi and Fransoo, 2025).
In general, the issue is not only that about 75% of the inland freight volume is transported via trucks, but also that these trucks contribute to a disproportionally high share of the emissions because they emit more emissions compared to other modes of transports (Fazi and Fransoo, 2025). These numbers show the importance of the industry for the economic situation of the EU, but the transportation industry also has a significant impact on the environment. As of 2024 the transportation sector is the biggest emitter of CO_2 equivalents with an absolute value of ~800 MT or about 28,8% of the total emissions in the EU (European Commission, 2025a; European Environment Agency, 2025a).
The EU Green Deal mandates a 90% reduction in transport-related emissions by 2050 compared to 1990 levels (European Commission, 2025b). However, since 2000 the transport emissions of CO_2 have remained steady and of these transport emissions 40% are associated to the freight transport industry (European Environment Agency, 2025b; Fazi and Fransoo, 2025). This shows that the progress made on lowering the emissions per vehicle of the last 20 years has just been enough to counteract the increasing number of transports happening in general.

Environmental impacts from the RFT often increase disproportionally compared to GDP Growth, further highlighting inefficiencies within the sector. One key contributor to this imbalance is the presence of a highly fragmented market. In the EU 91% of all carriers have less than 10 employees and this number is still skewed by a small number of very

big players that operate especially long-distance routes (Carvalho and Sousa, 2020; Grzelakowski, 2025). On the short distance routes where predominantly smaller carriers operate the majority of 85% of all carriers have fewer than 5 employees (Carvalho and Sousa, 2020). This fragmentation can also be seen in Sweden itself where the average number of employees is 4.7 per company (Statistikmyndigheten SCB, 2026). This fragmentation often leads to a highly competitive market where actors are seriously affected by external disruptions and low margins are common which in return limits the availability of capital to invest in new technologies. These factors have led to the market situation we are having right now. The transition happens slowly because carriers face the risk of bankruptcy in case they make a wrong decision (Carvalho and Sousa, 2020; Grzelakowski, 2025).

In addition to the fragmented and thereby tense market situation, the trucks are often not fully utilised or even empty on their trips which decreases their efficiency drastically. Current data shows that in the EU between 20 and 25% of all trucks drive empty to relocate the truck or the driver (Carvalho and Sousa, 2020; Larsson, Linåker and Runeson, 2024; Spini *et al.*, 2025). Empty running can in specific cases even reach dimensions of 50% because of seasonal differences in business or general market crises (Meßmer *et al.*, 2025). Moreover, even when trucks are not empty, they are rarely fully utilised. The average utilisation of trucks based on loading meters is estimated at around 57% which would leave about 43% of a truck space empty (Carvalho and Sousa, 2020; Meßmer *et al.*, 2025). When looking at the weight utilization, the numbers are even worse as the weight utilisation reaches average values of 22% (Meßmer *et al.*, 2025). These numbers are commonly referred to as the loading factor which is defined as the ratio or percentage of available space that is utilised. The most mentioned reasons for empty runs and low utilisation are inefficient route planning, lack of demand in certain areas and poor communication.

A way to deal with these empty runs is to foster collaboration between carriers, i.e. horizontal collaboration (HC) with the goal of combining freights and thereby increasing utilisation and reducing the environmental impact. The benefits of these collaborations have been evident for a while and a lot of research has been done on this (see chapter 2.2). However, despite the known advantages, the real-world application of HC remains limited to individual trials and no big scale collaboration has emerged yet. Current digital platforms (DP) facilitate basic interaction and especially selling additional capacity where the DPs act as a marketplace. However, the currently used DPs lack a full system integration which makes use of all the benefits of HC (Heinbach *et al.*, 2022; Fazi and Fransoo, 2025). A key barrier to the complete integration is not the lack of technology but the lack of trust between the carriers who fear a knowledge spillover and poaching of customers with the help of shared sensitive data (Lindawati *et al.*, 2015; Islam *et al.*,

2019). This fear is often greater than the foreseeable competitive advantage gained even though different technologies can minimize this data leakage (Agahari, Ofe and de Reuver, 2022).

Research Gap

While the simulations and modelling of HC have already shown that the logistic costs can be reduced by up to 47%, applications remain scarce due to the governance mechanisms required for data sharing between competitors (Aloui *et al.*, 2021). Different sources mention a gap between the theoretical optimal way of working and the practical implementation of these, the so-called design-reality gap (Masiero, 2016). However, several gaps remain underexplored.

The willingness to share data is for example often treated as an external factor rather than observing the conditions under which it is produced. Data sensitivity itself is often treated as a property of data categories without examining whether this assumption holds in contexts where the data recipient is a competitor. Lastly, while different governance models have existed for a long time in the Swedish context, no study has systematically examined its setup and the features of its architecture in regards to their effect on data shareability.

These gaps are not unique to the logistics sector alone but instead are present in various industries like the healthcare, education or the food industry (Jussen, Schweihoff and Möller, 2023; Jussen *et al.*, 2024; Gelhaar *et al.*, 2025). This study addresses these gaps through qualitative interviews with carriers and industry experts in the Swedish road freight market.

1.2 Purpose

The purpose of this study is to identify what conditions are required to make currently existing theories about shared data platform viable for HC for the RFT sector. It focuses on identifying the existing barriers that must be resolved for such a platform to succeed and gain traction in the industry. While the focus of the study itself will be put on the RFT sector, the gained results and insights will be used to develop findings that can be applied to the general phenomenon of platform adoption failure and data sharing in fragmented markets. The study aims to gain these qualitative insights from a combination of a literature review, carrier interviews and an analysis that is conducted in a primarily Swedish context.

1.3 Research Question

RQ1 What factors can explain the persistent gap between the theoretical potential of data-sharing platforms for horizontal collaboration in road freight transportation and their practical adoption among carriers?

- RQ1.1 What characterises the current state of data exchange and planning practices among carriers and to what extent do platforms fail to replicate this?
- RQ1.2 How do carriers in fragmented markets perceive and manage data-related risks when engaging in horizontal collaboration?
- RQ1.3 What relational and institutional settings shape carriers' willingness to participate and join data-sharing arrangements?

1.4 Limitations

This study has a set of constraints that define the boundaries of the study to allow for a critical assessment of the results in an academic context. The following points outline the limitations:

The empirical work is centred on road freight carriers whilst other stakeholders like shippers are only secondary actors. Carriers are prioritized because they bear the highest operational and competitive risk in data sharing and are also the actors where a platform could leverage its benefits the most. In addition, already now carriers are the actor that takes over most, if not all, of the route planning and therefore would also be the actor that would be using data-sharing platforms. Therefore, the results are focused on the requirements of carriers while other stakeholders are only mentioned briefly as part of the theoretical framework.

The study, while focussing on the general barriers and incentives to the adoption of data sharing platforms, has an empirical focus on the RFT sector. This focus aligns with the interests of VTI and allows for more in-depth research to identify underlying principles that may be transferable to the whole sector of collaboration through data sharing platforms.

A similar approach can be seen in the geographical focus of this study. While the literature study focuses on the broader EU to encompass general trends and have a broader theoretical foundation, the empirical work is specifically focused on the Swedish market. This prioritisation was made to limit the scope of the study and to be able to compare the different results to another. The Swedish market is a very fragmented one,

as seen before, whilst still being relatively open to new technologies that could increase efficiency and lower the environmental impact.

The aim of this study is to investigate the requirements and conditions for a DP to work in practice. A special focus will be laid on the intercultural relationship between trust and the necessary steps to foster a strong cooperation built on trust. The technical setup or the specific optimization algorithms that need to be developed to build this DP are not part of this study and will therefore not be discussed.

This research utilizes a qualitative case-based approach involving semi-structured interviews with 11 carriers and 3 experts. However, due to the limited sample size the study is only able to capture a general indication of the industry. The findings are therefore indicative rather than statistically generalizable and should be interpreted accordingly (see chapter 3 for a more detailed explanation).

The empirical nature of this study has the potential to introduce biases from both the participants and/or the author. This was tried to be reduced with measures further explained in the chapter 3.5 but is still a key point that needs to be considered.

1.5 Study Outline

The following section briefly outlines the structure of the study to give the reader an overview of the logical flow.

The first chapter established the basic context for the study and showed the important role the RFT plays in our globalized industry. The environmental impacts and increasing regulatory pressure were mentioned while also introducing the main inefficiencies that this industry faces, predominantly empty runs and low utilisation. At the End, the studies aim and the research questions were described.

The second chapter provides the necessary theoretical background by analysing the current state of the literature regarding the freight market, HC and data sharing in DP. It introduces three theoretical lenses that are applied throughout the analysis and the design-reality gap to connect the theoretical background to the empirical interviews.

Chapter three describes the research approach taken including the process used for the literature review and the process to conduct and analyse the semi-structured interviews. This chapter also mentions the measures taken to ensure academic rigour, quality and validity and states the ethical considerations that were taken.

The fourth chapter presents the empirical findings gained from the interviews with carriers and industry experts and structures them around the sub-questions.

Chapter five analyses and discusses the findings from the empirical interviews in relation to the research questions and the theoretical background. The aim is to identify patterns and insights that emerge from the study and apply them to the underlying challenges of data sharing platforms. This chapter also highlights the studies implications and suggestions for future research.

The last chapter concludes the study by summarizing the key insights and answering the research question.

2 Theoretical Background

This following chapter establishes the theoretical foundation of the study, by first explaining the current situation in the RFT sector, then introducing the relevant actors and examining the concept of HC before looking at the state of data sharing platforms in general and specifically freight data platforms. At the end, incentives and barriers actors currently experience will be explained with the help of three case studies.

2.1 RFT: Structural Fragmentation and the Design-Reality Gap

Road freight transportation describes the movement of goods on road networks in between different locations. It therefore acts as a crucial link for the global supply chains, that span across continents, because it provides flexible and direct transportation. The sector encompasses a wide range of activities ranging from long-distance hauling to last-mile delivery and transporting various different types of goods usually transported in standardized units (Samuelson, 2021; Fazi and Fransoo, 2025).

Actors:

These operations are performed or influenced by a set of stakeholders that will be described in the following:

- **Shipper:** They are often manufactures or suppliers that commission a transport. Their main goal is to ensure that the product reaches its destination on time, in the expected state and are cost-efficient as possible. For this reason, shippers are primarily motivated by economic considerations. It is common to either operate based on a long-term contractual commitment or arrange transports on a short notice through spot-markets. However, as soon as the goods leave the shipper, the responsibility transfers to the carrier, and the shipper no longer has the control over the freight.
  In recent years, there has been a trend in companies to move towards more sustainable practices with that aim of reducing their environmental emissions and shippers are even willing to pay a small premium for this (Fries *et al.*, 2010; Le Pira, Giuffria and Paddeu, 2024).

- **Carrier:** Carriers are the transport companies that actually perform the transportation. Carriers own their vehicles, employ the drivers and need to ensure

that they comply with all existing regulations. They are responsible for the daily planning and the dispatching of trucks which is a very complex task with multiple different variables to consider, also called the vehicle routing problem (Fazi and Fransoo, 2025; Spini *et al.*, 2025).

- **Lastbilscentral (LBC):** This is a concept that is prominent in the Swedish RFT sector and describes a cooperative organization of independent carriers that combine central functions in one single company. This company is often jointly owned by its members and allows carriers to profit from a larger scale to compete in bigger tenders. The central office, the so called centralen functions as a centralized unit that takes over functions such as transport planning, invoicing, marketing and customer acquisition and communicates with the individual carriers. These carriers may vary in size but can reach from as little as 1 employee up until more than 100 employees.

- **Consignee:** Consignees are the recipient of the physical goods and are manufacturers, retailers or individual consumers. Usually, they have to inspect the goods and accept the delivery before the transport is completed (Fazi and Fransoo, 2025).

- **Freight Forwarder:** A freight forwarder acts as an intermediary between shippers and carriers. There are different types of freight forwarders but in general they get information about shipments from shippers and then match a carrier that can complete this shipment (Fazi and Fransoo, 2025).

- **Policy-makers:** As the name already suggests, policy-makers are governing bodies that are responsible for designing and implementing a strategy that governs the road freight sector. They have different ways to intervene, such as regulations, incentives and investments. The transition to a more sustainable transport industry is currently the focus of policy-makers (Shoman *et al.*, 2023).

Types of Freight Consolidation:

Consolidation of shipment is the process of grouping multiple individual shipments to achieve a better capacity utilisation and thereby reduce transportation costs. In the transport industry two main consolidation types are commonly used and carriers often specialise on one of them.
Full Truck Loads (FTL) describes a freight where a transport is dedicated to a single

shipper, meaning that the shipper can utilise the whole space of the truck. This reduces the risk of damage or loss of goods and makes the transportation process a lot simpler as the shipper usually only has to drop the goods off at one location and have no intermediate stops. This type of transportation is more commonly used for longer distances. However, FTL are often more expensive for the shippers and the utilisation is often lower compared to LTL because the shippers do not have the exact amount of goods necessary to fill the whole truck (Wang and Kopfer, 2014; Heinbach *et al.*, 2022).

Less than Truckload (LTL) describes a transport where the freight of multiple shippers is consolidated into one truck. This has the benefits for the shippers that only need to pay for the space that they need and the system is very flexible. Especially in cases where the goods are transported from warehouses to smaller retailers or end customers, this approach is commonly used. LTLs can be very efficient if the carrier is able to fully load its truck, however if there are disrupting market conditions or the carriers are not able to fill the truck, the utilisation can be low. Lastly, LTLs require a lot more handling processes at the loading and unloading sites which often leads to the need for processing in warehouses and therefore increased lead times (Wang and Kopfer, 2014; Heinbach *et al.*, 2022).

Transport planning:

The transport planning is most of the time done by carriers themselves after they have succeeded in securing a transport. Transport planning is a vehicle routing problem that has the goal of minimising costs while considering high fixed costs from the vehicles, driver salaries and insurances, variable costs from fuel, tolls and maintenance, constraints from capacity limits, working hours and special needs of the cargo and agreements with the shippers regarding pick-up and drop-off within the agreed upon timeframes (Toth, 2014). Many smaller carriers still conduct this process manually or with basic software but because these carriers only have a small number of vehicles and routes, a better software would not necessarily improve the utilisation of their trucks. This isolated optimisation or decentralised planning often happens because carriers want to protect their data which leads to inefficiencies, as other actors are not aware of potential empty runs and can therefore not act upon it (Wang and Kopfer, 2014). In contrast, when larger carriers do their planning in a more structured way, the results are often a lot better based on the overall truck utilisation and the total amount of distance driven (Padmanabhan *et al.*, 2022). This is the exact reason why LBCs are gaining traction in Sweden as they allow the carriers to take advantage of the benefits of a combined planning process whilst keeping the actual operations and ownership at the carriers.

In general, it can also be said that the planning of FTLs is easier than planning LTLs because of the nature of LTLs that have multiple stops per trip compared to the one stop transport of a typical FTL. In addition, the type of good transported also plays a role for the planning. Cooled goods need to be transported in a refrigerated container while loose goods like gravel often require a dump trailer.

Current Challenges:

As global trade is expected to continue to grow by another 44% by 2050 whilst the modal split is expected to remain fairly constant, the RFT sector will continue to grow in the future, making cost-efficiency and sustainability more important than ever (Meßmer *et al.*, 2025). This growth can further increase the impact of existing challenges.

The EU is currently experiencing a significant lack of truck drivers. In 2023, 233.000 driver positions are unfilled and by 2029, 17% of Europe's current driver population will have retired further increasing the pressure (IRU, 2024). This shows that the issue is ongoing and might even get worse in the future. The main reasons for the shortage are regulatory hurdles, long and irregular working hours and a lack of recognition (Ji-Hyland and Allen, 2022). An improved utilisation of drivers time can help to relieve pressure on the existing workforce (Spini *et al.*, 2025).

Another challenge that the road freight transportation is facing is the high market fragmentation present in the industry. In 2022, 91% of freight carriers had less than 10 employees (Grzelakowski, 2025). This high fragmentation sparks price wars and lower the margins of everyone involved to less than 2% (Heinbach *et al.*, 2022). These low margins lead to a "capital-weak" industry that is missing key financial resources to transition their fleet to more sustainable vehicles that are often twice as expensive compared to their diesel counterpart (Grzelakowski, 2025). Additionally, the low financial buffer makes carriers highly susceptible to external disruptions. As a result, wrong investments are very threatening for the carriers, which can also be seen at the high level of bankruptcy which have reached a ten year high in 2025 (Manners-Bell, 2025).

The road freight transportation is by far the most used freight transport mode across Europe's main land but is also responsible for the highest total emissions of CO_2 among all transport modes across the EU (Fazi and Fransoo, 2025). In addition, the transport sector is the only sector where the emissions between 1990 and 2019 have increased (Spini *et al.*, 2025). This enormous number of emissions generated by one sector has led the EU to the introduction of the EU Green Deal that mandates a 90% reduction in transport related emissions by 2050 with a smaller goal in 2030 (European Commission,

2025b). In addition to this regulation there are other incentives to lower the emissions like the CO_2-based road toll and higher taxes on fossil fuels.

A key challenge that the RFT sector has to deal with is the low utilisation rate that is present in the industry right now. The average utilisation of trucks in the EU based on loading meters is estimated at around 57% which would leave about 43% of a truck space empty (Carvalho and Sousa, 2020; Meßmer *et al.*, 2025). In addition to the low utilisation rates, between 20 and 25% of the trucks even drive completely empty (Carvalho and Sousa, 2020; Larsson, Linåker and Runeson, 2024; Spini *et al.*, 2025). These opportunities represent €160 billion in potential additional revenue that are missed because trucks are not fully utilised (Carvalho and Sousa, 2020). But of course, it will be impossible to fully utilise all trucks because of flow imbalances, distances between the drop-off location and a following pick-up location, inefficient planning and a lack of information sharing. Some of the points are harder to improve one due to structural factors but the planning itself can be optimised and an improved communication in between carriers also seems like a possibility.

The mentioned challenges for the RFT sector are all interconnected and most of them can be linked to an inefficient utilisation. These low utilisation rates increase costs and contribute to additional emissions. At the same time increasing utilisation rates could:

1. Help to reduce the total numbers of trips which helps to lower emissions
2. Increase revenue per trip which should correspondingly improve margins
3. Lower the drivers working hours hence potentially relieve some pressure cause by the driver shortage

2.2 Horizontal Collaboration

The general definition of cooperation is the *"process of reaching objectives that cannot be achieved acting singly (or, at a minimum, cannot be reached efficiently). Collaboration includes all of the following elements: jointly developing and agreeing to a set of common goals and directions; sharing responsibility for obtaining those goals and working together to achieve those goals, using expertise of each collaborator*" (Krajewska and Kopfer, 2006).

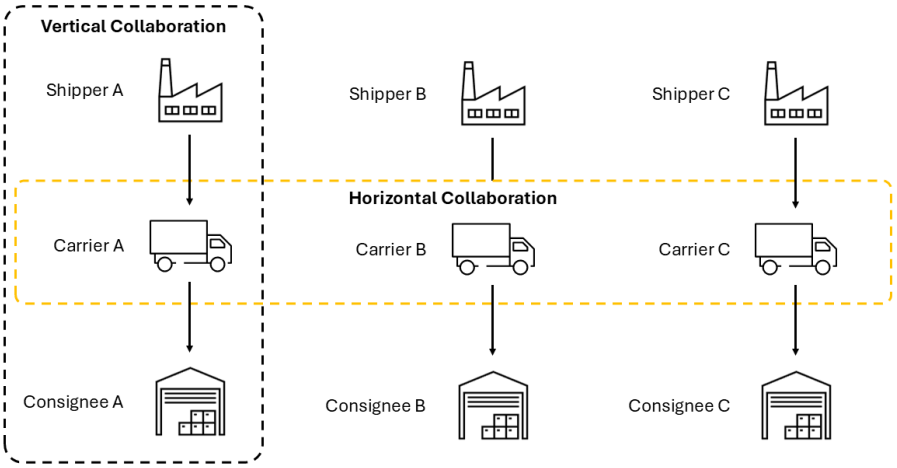


Figure 1: Horizontal vs. vertical collaboration

HC refers to the active cooperation between two or more actors that operate on the same level of the supply chain (Pan *et al.*, 2019; Aloui *et al.*, 2021). The opposite collaboration, vertical collaboration (VC), would include a partnership between actors on different levels of the supply chain, which could e.g. be a manufacturer and their suppliers (see Figure 1). While VC is a very common business practise, HC is less frequent as it involves a cooperation with an immediate competitor (Pan *et al.*, 2019; Mrabti, Hamani and Delahoche, 2022).

In the context of the RFT sector, the term horizontal collaborative transport (HCT) is also commonly used and can refer to the cooperation between all the different actors involved in this sector (Pan *et al.*, 2019). These collaborations have the common goal of reducing empty runs and increasing efficiency to increase the competitiveness (Cruijssen, Cools and Dullaert, 2007).

Pan et al. (2019) have developed a classification of HCT solutions that are grouped according to their scope and level of integration. There are a lot of different types of HCT but for the sake of this study only the two most prominent classes in Sweden are included as the others are deemed not fitting for the scope:

- Carrier alliance and coalition (S2): These groups exchange transport requests to mutually optimize transport across multiple fleets. While an alliance consists of independent units collaborating via decentralization, a coalition operates in a fully coordinated, centralized way, acting as a single integrated entity to achieve better global optimality. This means that carriers in an alliance might outsource routes or shipments that are not profitable for them, while actors in a coalition

combine their transports and optimize them centrally (Pan *et al.*, 2019). This way of cooperating can be seen in Sweden und the LBC.

- Transport Marketplaces (S3): These are digital platforms or marketplaces where shippers procure services and carriers can exchange transports flexibly, often utilizing auction mechanisms for spot-market agility. This so far is the typical scenario of a VC, but the transport marketplaces also allow other carriers to sell their transports, which resembles a HC. This system usually acts without a long-term commitment but therefore an actual cooperation is relatively rare (Pan *et al.*, 2019). This type of collaboration is also very important in the Swedish RFT sector. It is utilized by especially carriers working alone to sell additional capacity. This process is currently not digitalized and instead reliant on mailing lists and phone calls.

HCT is a concept that has been around for a longer time but has recently gained in relevance as the pressure on companies to operate more efficiently in the face of globalized competition rises. Benefits in the context of road freight transportation to collaborate, typically show up across economic, strategic and technical dimensions. A cooperation with direct competitors can allow both actors to profit from economies of scale. The primary driver for private stakeholders is profit maximization, typically achieved through reduced transport costs and increased sales by combining their LTLs in one vehicle (Le Pira, Giuffria and Paddeu, 2024). In addition, by combining the service offering and collaborating carriers are able to offer more services and routes which makes use of the economies of scope (Cruijssen, Cools and Dullaert, 2007; Pan *et al.*, 2019). The benefits of combining freights, eliminating resources that are now available twice and sharing them instead, and learning and sharing the strengths with a partner is a great way to profit financially from a collaboration (Cruijssen, Cools and Dullaert, 2007).
Shippers and carriers are also increasingly focused on reducing the "hidden costs" i.e. asset repositioning, where trucks move empty which collaboration can help to enable (Ergun, Kuyzu and Savelsbergh, 2007).
In addition, to the financial benefits, HCT is also a proven mechanism to decrease the environmental impact. There is a growing trend among private actors to align with public decarbonization strategies and the European Green Deal's goal of a 90% reduction in transport emissions by 2050 (Le Pira, Giuffria and Paddeu, 2024). A 10% reduction in trucking distance that can be achieved by combining freights, could achieve a $16 billion reduction in total global costs, while reducing the environmental impact at

the same time (Meßmer *et al.*, 2025; Mrabti, Hamani and Delahoche, 2022). Companies that collaborate can position themselves as innovative and sustainable, which serves as a key differentiating factor in highly competitive markets. Furthermore, being part of a growing ecosystem prevents firms from missing out on future scalability effects and brand awareness in the public (Jussen, Schweihoff and Möller, 2023).

Furthermore, HCT can improve service levels by increasing delivery frequency and geographical coverage, which reinforces the market position of the carriers (Cruijssen, Cools and Dullaert, 2007). This is especially valuable for small and medium-sized carriers, where HCT serves as a way to improve financial results and allows them to tender for larger contracts with industrial shippers (Wang and Kopfer, 2014).

Despite the shown and evident benefits, HCT has still only been implemented in specific cases and wide application is lacking. This is especially evident when seeing that about 80% of industrially generated data remains unused due to barriers that penetrate all levels of business and prevent the quick adoption (Jussen *et al.*, 2024).

The main reason mentioned in the literature for this lack of cooperation, is a lack of trust between carriers (Islam *et al.*, 2019; Karam, Reinau and Østergaard, 2021; Mrabti, Hamani and Delahoche, 2022). Carriers, functioning in a fragmented market with margins often below 2%, fear that sharing sensitive information like customer identities or route patterns will lead to "knowledge spillovers" or allow rivals to "poach" their clients (Lindawati *et al.*, 2015; Cruijssen, Cools and Dullaert, 2007; Grzelakowski, 2025). The literature however, makes a clear distinction between the different types of collaborations and their concerns in regard to trust. S2 alliance structures tend to succeed where members are non-overlapping geographically or by cargo type, enabling them to share capacity without directly jeopardising their respective customer bases (Cruijssen, Cools and Dullaert, 2007). S3 marketplaces, by contrast, expose carriers to a wider set of actors including direct competitors, which substantially amplifies the trust and information security requirements (Pan *et al.*, 2019). The governance arrangements that are unique to each model are therefore important structural prerequisites for the participation of carriers. This distinction can also help to explain the different responses to the same tool and why theoretical collaboration models do not reach the same adoption rate in reality.

A neutral platform provider or orchestrator to manage the collaboration process is also a way to overcome the "trust gap" that is mentioned in the literature (Pan *et al.*, 2019; Larsson, Linåker and Runeson, 2024). This third party can act as a trustee and provides a secure digital infrastructure where participants can exchange data and find synergies without revealing their core competitive intelligence to rivals (Jussen *et al.*, 2024;

Larsson, Linåker and Runeson, 2024). Such an impartial coordinator can help to ensure that information remains confidential, the distribution of work is fair, and the collaboration remains compliant with competition laws.

Another barrier that is often mentioned is the failure of sharing Information and Technology. This is often caused by a lack of advanced Information and Communication Technology systems, incompatible data formats and too high investment costs to implement different systems. For small and medium-sized companies, the additional effort of data pre-processing, cleaning before finally being able to start the analysis can be financially discouraging and requires technical skills that are simply often not available (Agahari, Ofe and de Reuver, 2022). One source even states that 70% of HC fails because of "*incompatible partners, information not collected efficiently, poor decision-making, and inadequate communication (Larsson, Linåker and Runeson, 2024)*"(Pan *et al.*, 2019; Karam, Reinau and Østergaard, 2021).

The selection of a partner is a crucial part where especially the personal fit must be given in combination with aligned business objectives. Ideally, the operational guidelines align to gain the biggest advantage from the cooperation, but finding this fit can be hard (Mrabti, Hamani and Delahoche, 2022).
Regulatory and legal concerns are prominent with smaller carriers that do not have the necessary legal expertise to be able to interpret regulations and avoid antitrust regulations (Pan *et al.*, 2019; Karam, Reinau and Østergaard, 2021; Larsson, Linåker and Runeson, 2024). In a similar way, concerns about the sovereignty of data and the fear of violating laws such as the General Data Protection Regulation, act as significant deterrents (Jussen *et al.*, 2024; Peet, Bharosa and Janssen, 2025).

A last barrier to a successful HCT initiative could be the compensation model. A fair mechanism for distributing the savings or common costs among participants must be ensured to prevent the breakdown of even the most efficient collaborations (Frisk *et al.*, 2006; Aloui *et al.*, 2021; Karam, Reinau and Østergaard, 2021). However existing optimisation methods like the so-called Proportional Approach and the Cooperative Game Theory can already now bridge this gap.

2.3 Socio-Technical Dynamics

As seen in chapter 2.2, HC has the potential to improve the utilisation and reduce inefficiencies in the RFT sector. These collaborations however have some barriers to them which often prevent the formation of these HC. Throughout the years data sharing

platforms have been changed and adapted to meet the customers' needs. The following section therefore introduces the general concepts of inter-organizational data sharing and socio-technical dynamics before relating their specific role to the RFT sector.

2.3.1 Concepts and Theories of Data Sharing Platforms

The evolution of data from a byproduct to a strategic organizational resource has fundamentally altered the landscape of modern businesses. In recent years, the volume of data has seen exponential growth, rising from approximately 104 zettabytes in 2022 to over 132 zettabytes in 2023 (Möller *et al.*, 2024; Pampus *et al.*, 2025). This rapid increase of the amount of data has shifted the focus of managers towards the creation of value through inter-organizational data sharing, a process defined as making data available to third parties for purposes of optimization, innovation, or value co-creation (Jussen *et al.*, 2024).

Current research distinguishes between traditional, sequential data sharing in supply chains and the emergence of Industrial Data Ecosystems (Möller *et al.*, 2024). While supply chain sharing is typically bilateral and aimed at mitigating effects like the bullwhip effect, data ecosystems represent complex networks where actors interact to find, consume, and reuse data to support different business models. These ecosystems are often facilitated by data intermediaries, such as data marketplaces or data trusts, which act as neutral platforms to coordinate the exchange and are therefore interesting to the RFT sector.

These Industrial Ecosystems are a promising concept that could drastically change the understanding in the RFT sector. However, the adoption of these systems is coherently dependent on the social-relational architecture through which the sector currently operates.

In the RFT, the dominant mode of coordination operates through knowledge-based trust (trust developed through repetitive interaction) and identification-based trust (trust developed because of a shared identity) (Lewicki and Bunker, 1994). Carriers build long-term bilateral relationships where dispatchers know each other by name and have accumulated years of reciprocal favours. This structure enables a degree of flexibility and mutual adaptation that formal contractual structures cannot replicate at comparable cost (Larsson, Linåker and Runeson, 2024; Grzelakowski, 2025). This flexibility dimension is not as closely observed and analysed in the HC literature event though it plays a crucial role in its adoption. Current carrier coordination is resilient because it relies on human judgment embedded in long-term relationships. Dispatchers carry significant tacit knowledge i.e. which unloading ramp is awkward to reverse into and therefore needs more time, which partner can be trusted for a same-day emergency load,

or which roads are inaccessible on specific days. This knowledge is not stored in any transport management system (TMS) and cannot easily be extracted into a platform.

Empirical evidence from the Swedish market suggests that the technical categorization alone fails to explain the significant design-reality gap (see chapter 202.3.3). To understand why certain platforms, succeed while others do not, it is necessary to transition from a technical typology to a classification based on governance archetypes and the social-relational foundations they require.

2.3.2 Actor-Network Theory and Privacy Calculus

The willingness of an organization to participate and join these collaborating platforms can be viewed and explained through these two theoretical frameworks that are applied as analytical lenses in the empirical analysis that follows in chapter 5. The key takeaway is that the adoption of this is not only a technical challenge but a complex socio-technical issue due to the underlying system dependencies (Jussen *et al.*, 2024):

- Actor-Network Theory (ANT): This theory explains the Translation Process through which a data-sharing network is formed. It involves four phases: identifying the problem (Problematisation), convincing partners (Interessement), operationalizing roles and infrastructure (Enrolment), and scaling the ecosystem (Mobilisation). This translation process is often a big challenge especially in systems like the RFT sector. Here systematic dependencies from the past like the dependency on phone calls and tacit knowledge still have a big impact on the way of working and make it hard for companies to digitalize. A successful translation often requires one actor to take initiative that promotes the idea to increase industrial acceptance and this is often done through an external discrepancy (Jussen *et al.*, 2024). What the ANT framework underspecifies, however, is the role of governance structure in enabling or blocking the translation process.

  In the Swedish carrier market, the translation process has already been completed for actors that have joined an LBC because the governance structure of them has pre-resolved the competitive conflict that otherwise would makes Interessement and enrolment impossible. For carriers outside such structures, the translation process stalls at the Interessement stage because there is no mechanism that can promise them that their data will not be commercially exploited. The platform's value proposition is thus conditional on a governance prerequisite that most current platform designs fail to provide.

- Privacy Calculus Theory: The Privacy Calculus theory analyses how firms balance the perceived benefits (incentives) against the perceived risks (barriers) before making a decision on sharing data. This concept was applied to the topic of data disclosure by Dinev and Hart (2006) who introduced the two opposing classes risk beliefs and confidence/enticement beliefs. Their findings demonstrate that these two sides are not symmetric, but that trust in the recipient has a stronger effect that the inhibiting effect of privacy. Their conclusion is that rather than reducing the perceived risk, the credibility of the recipient can also be increased. In the RFT sector the perceived risk ranges from data misuse to a fundamental lack of trust in the platform coordinator. These barriers are very present in this market but also fragmented markets in general, which often results in a negative balance regarding data sharing (Jussen, Schweihoff and Möller, 2023). Dinev and Harts (2006) framework implies that this negative balance could be neutralised or even tipped when a governance agreement regarding data sharing is established.

To mitigate these risks, emerging technologies like Secure Multi-Party Computation are being integrated into platforms. This allows multiple parties to jointly compute a function over their inputs while keeping those inputs private (Agahari, Ofe and de Reuver, 2022). In an industrial context, this means a platform can provide direct insights instead of raw data which allows competitors to collaborate on shared problems without revealing sensitive business secrets.

2.3.3 Patterns of Data-Sharing Governance

The explained difference between the design and reality fits the concept of a design-reality gap. This concept, initially developed in the information systems literature to explain the differences between the design and the organisational reality, can help to explain the adoption failure in the RFT sector (Masiero, 2016). The gap describes to what extent the assumptions used in a system deviate from the actual conditions of the environment in which it is used. These assumptions can in reality reach from the available infrastructure to the organisational processes in place. A system tends to fail when any one of these dimensions diverges too far from the design idea or when multiple dimensions are misaligned. The resulting gap combines making adoption more difficult. Larsson et al. (2024) apply this framework to the Swedish RFT context, demonstrating that failed digitalisation in this sector consistently shares a structural pattern namely that platforms are designed assuming a level of data standardisation, partner interchangeability and willingness to disclose information that does not reflect the

reality. The occurrence of this mismatch across multiple dimensions makes the failure difficult to resolve.

Building on the previous section about the design reality gap, the following part focusses on platforms that are built for the RFT sector. The current market is characterized by a wide array of technical solutions aiming to solve market fragmentation and operational inefficiencies. In the Literature these DP are often classified by their functional utility. Heinbach et al. (2022) have established four main groups of DPs namely:

- Freight Order Coordination (e.g., matching supply and demand)
- Freight Resource Handling (e.g., freight exchanges)
- Transport Data Connectivity (e.g., telematics integrators)
- Transport Process Support (e.g., workflow management)

Each category implies different data requirements and different degrees of exposure to competitive risk. The freight marketplace (S3) falls within the first two categories and requires disclosure of capacity, route, and pricing data to an audience that typically includes direct competitors. This is precisely the data that carriers guard most carefully, and the mismatch between what the platform requires and what carriers are willing to disclose constitutes the core design-reality gap in this context. As seen in Table 1, the market is currently dominated by platforms focusing on matching shippers to carriers (VC) instead of carrier-to-carrier collaboration (HC).

Platform	Freight Order Coordination	Freight Resource Handling	Transport Data Connectivity	Transport Process Support
UTURN uturn-now.com		X		
Flexport flexport.com	X			
C.H. Robinson chrobinson.com	X			X
Sennder sennder.com	X			

Transporeon transporeon.com		X	X	X
Timocom timocom.se		X		

Table 1: Classification of digital RFT platforms

A significant limitation of these platforms is that they treat transport as a service with flexible lead times, overlooking the structured and scheduled nature of most freight. Consequently, services are often only offered by large digital forwarders who can manually handle the complexity that automated matching algorithms currently lack (Fazi and Fransoo, 2025).

To address these gaps, researchers propose the concept of Freight Mobility as a Service. It is envisioned as an open, user-centric booking platform where shippers have full visibility of the global transport offer across all transport modes and would book a transport in a similar way as a passenger would buy a bus transportation. Unlike traditional platforms where decision-making is in the hands of transport operators, Freight Mobility as a Service allows shippers to autonomously select services based on requirements for time, budget and sustainability (Fazi and Fransoo, 2025).
However, the practical implementation of Freight Mobility as a Service faces significant hurdles. Adoption barriers remain high as stakeholders are reluctant to change traditional booking methods or increase transparency. Additionally, implementing a change along the whole SC is a very big step which further increases resilience form stakeholder involved. Finally, the incentive structures for carriers to join such open ecosystems remain a critical area and could be explored further in the future (Fazi and Fransoo, 2025).

2.4 Existing Collaborations

Inter-organizational data sharing in the RFT sector is a complex balancing act where actors must navigate a decision process influenced by both technical capabilities and deep-seated social norms. To understand the existing design-reality gap, this section introduces organizational models and cross-industry cases that have attempted to bridge the tension between individual competition and collective efficiency.

2.4.1 The Swedish Lastbilscentral

The LBC model represents a unique institutional architecture that differentiates itself from the typical operational models of a carrier. The setup used avoids data asymmetry by aligning the ownership of the entity with the data contributors (carriers) themselves.

Historically, LBCs emerged as cooperatives of small, independent carriers seeking the economies of scale typically reserved for large-scale integrators. Unlike a third-party digital platform, the centralen is co-owned by the participating carriers, which transforms the act of sharing from an external disclosure to an internal administrative function. This dual role transforms the nature of data disclosure: providing operational information to the centralen is not perceived as sharing "strategic positioning data" with a rival, but rather as contributing to a shared administrative infrastructure that the hauliers collectively govern. The LBC acts as a neutral intermediary that manages centralized planning, dispatch and the commercial interface with large industrial shippers which small carriers themselves could not efficiently handle alone.

The success of the LBC model illustrates that horizontal data sharing is achievable when the Privacy Calculus is pre-resolved. Because the governance structure eliminates the threat of "customer poaching" from within the organization, carriers are willing to relinquish operational autonomy in exchange for the aggregated efficiency and market access provided by the collective. This model suggests that the primary barrier to digital platform adoption is not a lack of technology, but a lack of institutionalized trust that protects the carrier's competitive advantage. LBCs can therefore serve as a proof of concept for this study. They demonstrate that horizontal data sharing is achievable when the conditions are met.

2.4.2 Strategic Coalition "OnRoad"

Building upon the LBCs, the OnRoad network serves as an example of HC across regional boundaries. OnRoad is a coalition of LBC-affiliated carriers that jointly manage cross-regional transport flows throughout Sweden and support each other in case of missing capacity (*OnRoad - National transport Network*, 2026).

The network provides an indication that data sharing can scale horizontally when participants operate in different geographic areas. By aligning these carriers, OnRoad minimizes the risk of internal competition that typically deter collaboration. The network succeeds because its governance rules are established in advance, creating a shared institutional identity that facilitates the "enrolment" and "mobilisation" phases described in the Actor-Network Theory (see chapter 2.3.2).

However, even within such successful coalitions, digitalisation and the slow integration of systems are a reoccurring issue. Even though digital systems can manage the bulk of order routing, the urgent deliveries and any transport that has deviations are still resolved through manual communication between dispatchers. This shows the clear dependency on human judgement and knowledge which could be a barrier to the platform adoption as they currently fail to replicate these skills.

2.4.3 Cross Industry Case Studies

The challenges of data sharing in road freight are not unique to the logistics sector but are systemic across fragmented industries where independent actors compete on thin margins. In the following section three short examples of this reoccurring phenomenon will be shortly introduced.

The first short case study deals with a Norwegian Fishing Fleet (Sønvisen *et al.*, 2025). The fishing sector provides a parallel to the fragmented road freight market, where independent vessel owners operate with high operational risks. Findings revealed a widespread reluctance to share catch data due to fears of losing a competitive edge on specific fishing grounds, which is a similar situation to road carriers. In addition, Fishers expressed suspicion that data-sharing technology serves the economic interests of equipment suppliers rather than the operators. Lastly, the study identifies trust and access to data as critical balancing factors. Fishers were only willing to share detailed data if it is anonymized or subject to a "publication embargo" to protect their business secrets.

The second case study is about a Standard Business Reporting in the Netherlands (Peet, Bharosa and Janssen, 2025). This case study examines a multi-actor trust framework designed to reduce administrative burdens on businesses through automated data exchange with public agencies. The Standard Business Reporting initiative illustrates how institution-based trust can be built through formal agreements and inclusive governance where all stakeholders, including smaller software providers and intermediaries have a voice and are heard. By using standardized data formats and rules to ensure data accuracy, the framework successfully mitigated concerns about data quality and authenticity. This case highlights that trust does not emerge by chance but is the result of an intentional design, which can be linked backed to the Actor-Network Theory.

The last case mentioned here deals with Shared-Fleet Logistics in the public healthcare sector (Grote *et al.*, 2023). A real-world trial involving a Patient Transport Service and a Sample Collection Service demonstrated that sharing spare vehicle capacity could reduce costs by 16% and CO_2 emissions by 12%. However, even in this non-competitive public

setting, significant barriers were identified that included vehicle delays and service level agreements that needed to be met. The study concludes that while the theoretical benefits are high, practical realities such as liability for damages and the reluctance to use alternative providers require extensive negotiation before implementation.

The three cases illustrate how data sharing can help processes in real scenarios but also show that problems must be expected and dealt with accordingly to keep a good and beneficial collaboration. In addition, it showed the importance of trust and the applicability of structures across sectors as there are similar concerns and key considerations for all cases.

The literature reviewed in this chapter establishes a theoretical foundation for this study. It is structured around three analytical lenses contributing to a distinct perspective on the problem of digital platforms that this study investigates. The first lens, the design-reality gap by Masiero (2016) and applied to the Swedish RFT context by Larsson, Linåker and Runeson (2024), deals with the recurring pattern in which DPs fail. The Privacy Calculus framework of Dinev and Hart (2006) provides a theory that examines how individual carriers weigh the benefits and risks of sharing data against each other and how the credibility of the recipient can affect this. The ANT by Jussen et al. (2024), describes the process through which a data-sharing network is formed and either succeeds or identifies where the failure occurs.
These three lenses serve different functions in this study. The Privacy Calculus framework is the primary analytical lens through which the adoption question is examined. The empirical analysis aims to test and refine this framework while the design-reality gap framework and Actor-Network Theory function as a guideline. Together they frame the empirical questions: what does the carrier's cost-benefit calculation actually look like in this market, what governance conditions would change it and at which stage of the translation process does platform adoption stall? The case studies and existing collaborations in chapter 2.4. complement this picture by showing that the barriers are not unique to Swedish road freight. These observations guide both the interview design and the analysis that follows.

The literature points the adoption failure in the RFT sector to a mismatch between the conditions that actors require and those available in the market. This misalignment guides both the research questions and the empirical study that follows.

3 Methods

Chapter 3 outlines the methodological approach applied in this study. It begins by presenting the research design, the research process and then the steps taken for the literature review. Then follows a description of the data collection process and the data analysis methods, as well as key considerations to ensure research quality and ethics.

3.1 Research Design

The nature of this study is identified as exploratory, as it aims to gain new insights into the requirements and conditions for shared data platforms to become viable for HC in the road freight transportation. By selecting this approach, it is important to be able to adopt the focus rather than being fixated on a specific result (Saunders, Lewis and Thornhill, 2019). Exploratory research is particularly well-suited to situations where the phenomenon of interest is complex and where the boundaries of the study cannot be fully specified in advance. All of this characterises the present study, where the interplay between structural fragmentation, social trust, and platform design is poorly understood in the specific Swedish RFT context (Yin, 2018).

Aligned with the qualitative research choice being recommended for exploring interactions between actors, the study employs a case study research strategy. This strategy allows for an in-depth investigation of the Swedish RFT sector in a real-life context where the boundaries between data-sharing events and social barriers are not always clear. Yin (2018) argues that a case study is the preferred strategy when "how" and "why" questions are being posed, when the investigator has little control over events, and when the focus is on a contemporary phenomenon within a real-life context. These conditions apply directly to this study. Moreover, Eisenhardt (1989) contends that case study research is especially powerful for building theory in areas where existing theory is insufficient or where dominant theoretical frameworks have not yet been tested in a specific empirical setting, which is precisely the situation encountered here with respect to horizontal data sharing in fragmented freight markets.

The study follows an abductive approach to theory development, enabling an iterative process that moves back and forth between existing literature on data ecosystems and the primary empirical data. This is consistent with what Dubois and Gadde (2002) describe as "systematic combining," a process in which the theoretical framework, the empirical observations and the evolving analysis are developed simultaneously and in mutual interaction. The abductive reasoning allows to return repeatedly to existing

theory when confronted with empirical observations that cannot be explained, using those observations as a lever to refine and extend the theoretical framework (Dubois and Gadde, 2002). This allows for a final study that has established a close link between the literature and the empirical data which helps with the conciseness.

The Data collection itself is conducted through semi-structured one-on-one interviews, which are particularly effective for answering the explorative questions central to this study and fit well with the previously mentioned explorative study (Saunders, Lewis and Thornhill, 2019). Semi-structured interviews provide the flexibility needed to understand the emerging themes in depth while still maintaining sufficient comparability across respondents to enable systematic cross-case analysis (Brinkmann and Kvale, 2015). This is particularly important given that the sample spans both operational carriers and higher-level industry experts, whose perspectives on data sharing differ substantially in orientation and abstraction.

3.2 Research Process

The research process for this study followed an iterative logic, allowing the research to continuously adapt as new insights were uncovered during the data collection process, an approach promoted by Saunders, Lewis and Thornhill (2019) for exploratory research. The process was here for structured into three main phases that will be shortly mentioned:

The initial phase involved a literature review to establish a theoretical foundation for this study. This step is crucial for identifying the research gap and to determine the key barriers that are to be explored. As the abductive approach requires, the literature was not treated as a fixed framework imposed on the data but rather as a dynamic resource consulted and revisited as the empirical work progressed.

The second phase focussed on the data-collection and structuring of the data collected through semi-structured interviews with carriers and industry experts. This phase was an adaptive process which means that the interview guidelines were adapted throughout the process to adapt it to the specific requirements. The adaptive process also helped to deeper explore new topics that emerged from previous conversations. In addition, the recorded interviews were transcribed and coded to easily spot reoccurring topics and relationships between the interviews.

The third and last phase of the research process was the interpretation and discussion of the results gained. During this stage, the empirical findings were compared with existing literature to highlight consistencies, inconsistencies, and potential new insights. This

comparison served as the basis for answering the research questions and providing practical implications for the development of shared data ecosystems as a whole.

3.3 Literature Review

The literature review for this study was conducted to establish a theoretical foundation that combines the developments and requirements of the RFT sector whilst at the same time capturing the developments in the sector of data-sharing platform. Managing these different topics which encompassed logistics, digital governance and trust frameworks was done with the help of Zotero, where the literature was clustered into thematic streams.

The review itself was started by generating a research string and using predominantly Scopus as a database, which yielded good initial results which were filtered by academic relevance and rigorousness. The results were initially scanned by abstract before being read in more detail which then introduced new relevant articles through a snowballing process to expand the scope of the search. The AI research tool ResearchRabbit was utilised to see the connections between papers and also find other papers that have not been included in the search before but seemed highly relevant.

This iterative methodology ensured that the review moved beyond a linear summary and allowed for an adaptive approach that ultimately led to changes in the selection of the research stream to contribute to.

3.4 Qualitative Study

The qualitative part of this study is based on semi-structured interviews that were conducted according to Saunders, Lewis and Thornhill (2019) and guided by a set of questions focussed on the study and adapted throughout the process. The choice of a qualitative approach is based on the assumption that the barriers to platform adoption in the RFT sector are fundamentally social and therefore cannot be adequately captured through quantitative metrics alone. As Yin (2018) notes, qualitative methods are irreplaceable when the aim is to understand how and why actors behave as they do within a specific institutional and relational context.

3.4.1 Data Collection

A total of 14 semi-structured interviews were conducted with relevant actors from the RFT sectors. The interviewees were primarily selected based on Swedish road freight

carriers and, in a smaller number of cases, industry experts with experience in digital collaboration and/or working with carriers. Relevant carriers were selected with the help of “allabolag” (https://www.allabolag.se/), Fair Transport (https://fairtransport.se/) and regular google searches and keywords such as: Åkeri, Lastbilcentral, Spedition.
The split into carriers and experts was made to capture a broad spectrum of perspectives from both operational and strategic levels.

Before the initial interviews were conducted, an interview guide was developed based on the key themes identified in the literature (see Appendix - Interview Guidelines). The interviews followed a similar approach with open-ended questions to allow participants to share their views freely, supplemented by targeted questions around specific categories derived from the theoretical framework. All interviews were conducted with the explicit approval of participants for audio recording and subsequent transcription, providing a basis for the coding and analysis that followed. The recording and immediate transcription of interviews is widely regarded as a prerequisite for rigorous qualitative analysis as it minimises the risk of data distortion and ensures that informant language is preserved as closely as possible (Miles, Huberman and Saldaña, 2014).

The interviewed people and a description of the role can be seen in Table 2. The interviews 1, 2 and 3 were the mentioned Expert interviews.

#	Position	Years of experience	Company Size	Interview Duration	Date conducted
1	Researcher	4	n/a	69 min	08.04.2026
2	Head of Operations Europe	14	n/a	55 min	14.04.2026
3	Founder and Consultant	30	n/a	68 min	16.04.2026
4	Chief of Logistics	12	100-500	52 min	01.04.2026
5	Head of Transport department	15	100-500	62 min	08.04.2026
6	Senior Commercial Manager	11	100-500	47 min	08.04.2026

7	Sustainability Manager	19	100-500	60 min	09.04.2026
8	Chief of Logistics	21	100-500	54 min	10.04.2026
9	CEO	32	10-50	46 min	10.04.2026
10	Head of Transport department	16	500-1000	44 min	13.04.2026
11	Managing Directors Hauliers	23	500-1000	57 min	13.04.2026
12	Operations Manager	4	100-500	51 min	15.04.2026
13	Sustainability Manager	10	10-50	42 min	20.04.2026
14	Chief of Logistics	18	100-500	67 min	21.04.2026

Table 2: List of conducted interviews

3.4.2 Data Analysis

The data analysis for this study is conducted using a systematic thematic approach, primarily guided by the Gioia methodology to ensure qualitative rigor (Gioia, Corley and Hamilton, 2013). This method was explicitly designed for research in which the aim is to understand phenomena as actors themselves understand them, and has since become a widely applied and peer-reviewed framework for inductive and abductive qualitative research in management and organisation studies (Gehman *et al.*, 2018).
This process is also iterative, just as the literature review, allowing themes to emerge directly from the empirical interview data rather than forcing them into predefined hypotheses. This aligns well with the exploratory nature of this study.

The Gioia methodology builds on a social constructionist perspective informants themselves use to make sense of their world before translating these into progressively more abstract theoretical categories. This approach prevents against the premature imposition of theoretical categories on the data and ensures that the resulting analytical framework maintains a traceable and auditable connection to the empirical reality from which it emerges (Corley and Gioia, 2004).

The analysis follows a structured three-stage coding process that moves progressively from raw empirical material to theoretical dimensions. In the first stage, open coding is applied to the transcribed interview material, in which verbal statements and close paraphrases of informant language are broken into descriptive codes. At this stage no categorisation or theoretical interpretation is done meaning that the statements keep their "informant-centric terms" (Gioia, Corley and Hamilton, 2013). The goal here is to generate a comprehensive and inclusive set of first-order concepts that capture the full range of opinions, experiences and concerns expressed across the interviews before any abstraction takes place. Miles, Huberman and Saldaña (2014) similarly emphasise that this initial phase of data reduction must remain descriptive and close to the data in order to avoid collapsing potentially important distinctions too early in the analytical process.

In the second stage, the similarities and differences across first-order concepts are systematically examined and a more abstract level of coding is applied to related codes into second-order themes. Whereas first-order concepts are descriptive and close to the informants' own language, second-order themes are interpretive and begin to reflect the researcher's theoretical understanding of what the recurring patterns in the data signify (Gioia, Corley and Hamilton, 2013). The process from first-order concepts to second-order themes is itself iterative and as new interviews are completed codes may be reconsidered, merged or renamed. This iterative approach is consistent with the abductive logic of the broader research design (Dubois and Gadde, 2002) and is described by Miles, Huberman and Saldaña (2014) as an indispensable feature of rigorous qualitative data reduction, preventing premature closure around categories that are convenient but theoretically imprecise.

The third and final stage abstracts the second-order themes further into aggregate dimensions that encompass the theoretical framework of the study. Aggregate dimensions represent the highest level of abstraction in the Gioia data structure and are designed to connect the empirically grounded themes directly to the conceptual contribution the study seeks to make (Corley and Gioia, 2004; Gehman *et al.*, 2018).

In the present study, the aggregate dimensions are organised in three main categories. These dimensions encompass the topic of data usage and exposure, the connection between trust and governance structures, and the existing gap between user needs and platform deliverables.

The resulting data structure, including the first-order concepts, second-order themes and aggregate dimensions, is visually presented in Figure 2, following the conventions established in the Gioia methodology literature. This visualisation transparently shows the connection between the first-order concepts and aggregate dimensions, which is central to ensuring the confirmability of the findings (Gioia, Corley and Hamilton, 2013).

In addition, an example of the coding for selected first-order concepts can be found in the Appendix – Interview Codes.

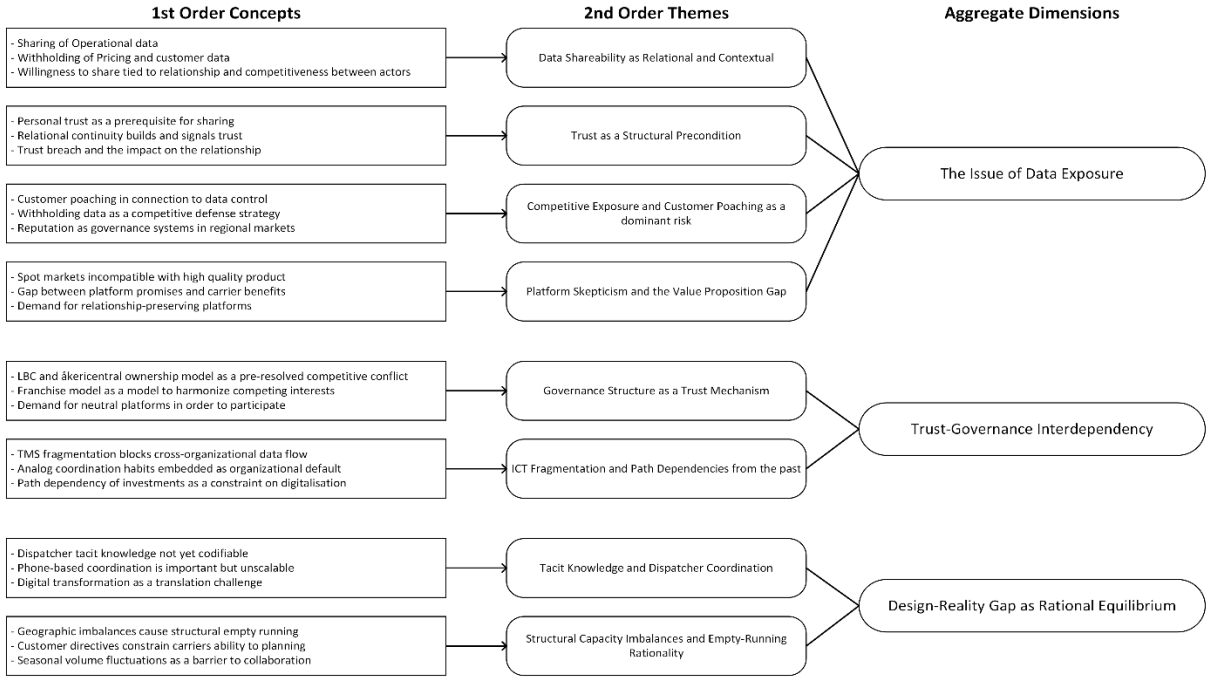


Figure 2: Gioia Interview Coding

3.5 Research Quality

In order to ensure high ethical research standards it is essential to persuade the reader of the trustworthiness of the findings (Lincoln and Guba, 1985). According to them it is important to address the following four criteria:

- **Credibility:** Credibility is ensured through triangulation by comparing literature findings, interview data and feedback from peers and supervisors with each other. These pillars ensure that the results can be seen as credible. Triangulation across data sources and interpretive peer debriefing are the most commonly recommended strategies for enhancing credibility and both are employed here through the combination of primary interview data with secondary literature and continuous feedback from both the academic supervisor and the VTI research partners (Saunders, Lewis and Thornhill, 2019).

- **Dependability:** Dependability was ensured through a transparent research process with external reviews from VTI, peers and the academic supervisor. Furthermore, the roles of the interviewees are disclosed whilst ensuring

anonymity to further support the reliability. The step-by-step documentation of the research process in this chapter, the availability of the interview guide in the appendix and the explicit description of the coding procedure all contribute to dependability.

- **Transferability:** This study addresses external validity by extracting the key findings and applying it to other sector that struggle with trust in data sharing environments. While this study is deliberately focused on the Swedish RFT context, the theoretical framework built on concepts such as the privacy calculus, the design-reality gap and ecosystem governance is constructed at a level of abstraction that has clear parallels other fragmented markets facing similar barriers. This supports a degree of analytical generalisability, even if statistical generalisation is not the ambition of this study (Yin, 2018).

- **Confirmability:** To ensure a minimized bias from the author, the Gioia methodology is used to code the interviews. This visualization of the data with links from raw statements to the dimensions provides a rigor connection between data and emerged theory which is confirmable (Gioia, Corley and Hamilton, 2013). By showing the process in the appendix and documenting the reasoning behind each coding decision, the analysis is transparent for an independent reader to assess whether the interpretations are well-founded.

In addition to ensuring methodological rigorousness, this study followed ethical principles for social research, including the requirements of informed consent, confidentiality, and responsible data use (Saunders, Lewis and Thornhill, 2019). All participants were informed about the purpose of the study, their voluntary participation and the right to withdraw from the study at any time after the interview was conducted. Before the interviews were conducted approval for audio recording and transcription was established and the collected data was anonymized to an extent that it cannot be traced back to individuals. Lastly, the recordings were deleted after transcription to protect participant privacy.

Finally, the use of external AI tools was clearly stated and these tools were not used to actively write any new text. The exact type of usage can be found in the Declaration of the use of AI at the beginning of the study.

4 Results

The following chapter presents the empirical findings derived from the thematic analysis of fourteen semi-structured interviews conducted with Swedish road freight carriers and industry experts. The analysis follows the Gioia methodology outlined in chapter 3. The chapter is organized around the three sub-research questions guiding this study, but the sections here reflect the empirical logic through which the existing underutilisation of digital data-sharing platforms in the RFT sector can be explained.

4.1 Current State of Data Sharing

Before it is possible to understand why carriers do not share data on digital platforms currently, it is necessary to understand how they store, use and handle data at all. The empirical picture that emerges across the interviews is one of a sector in which digital tools exist, but in which cooperation between organisations largely does not. Carriers operate with a range of TMSs which often do not have an integration with others and the dominant mode of communication remains the telephone call or emails.

> *"The digital maturity in the logistics sector is stuck in 1999. Basically they have computerized the system, but they have not gone digital. The processes are still in many segments based on the analogue precursors to them that were developed before 1999."* (Interview 3)

> "*This [planning] is almost always email or phone. That's all. You have email lists and then you just put it [the transport] out and everybody knows.*" (Interview 9)

This distinction between computerization and digitalization with a particular focus on the correct implementation is important and something that has not been carefully done in the past. Computerization itself describes the digitizing of analogue processes, genuine digitalization requires restructuring those processes and adapting them to capture the capabilities of the used system. On this measure, most carriers remain in the computerization step. This assessment is confirmed from a practical example:

> *"It is quite manually with I would say mainly with Google Maps to see routes and count the kilometres."* (Interview 10)

However, certain carriers have an ongoing transition even though this process is only partial and a step-by-step approach is taken. The characterization is of a sector in which digital tools have been layered on top of existing analogue processes rather than used to redesign those processes from the ground up.

> *"I would say that we've not been very good at in in the IT and communication because we have a mix of different ways to communicate to the driver. […] So now you assign the driver a drive and he is able to go into an app and see okay I have to do this with this truck and so on. It's a step to make it more transparent so the driver will know what to do. Because until now, we've been sending them load lists"* (Interview 11)

The structural cause of this pattern is not primarily technological reluctance but the way the system has developed and what it was built on. The legacy infrastructure problem is therefore imminent and independent of the scale of the company even though bigger companies are more willing to try out new technologies. This matters for the assessment of platform adoption and data-sharing because any platform that requires a significant standardisation and integration between systems as a precondition is challenged with the current state of the industry.

> *"One of them is the path dependency that we have the systems we have. It's expensive to just replace them just because something new comes along. And the bigger the organization, the bigger the tech debt that you have. […] DHL has not [gone digital]. They have a computerized system, not a digital one."* (Interview 3)

The coordination and data sharing that does occur between carriers mostly happens through bilateral personal relationships. Interview person 8 describes the coordination with other carriers as a *"contact network rather than integrated data exchange"*. This interpersonal architecture is robust and very flexible, which can be of good use in this industry, but the system is not able to scale in that way. The dependency on the knowledge of individual persons rather than on technical infrastructure prevents the scalability that a full integration can offer.

The characterization consistently states that digital tools are being introduced at the interface without changing the underlying coordination logic. The structural cause of this pattern is not primarily technological reluctance but the accumulated cost of replacing systems that the organization has built itself around. Replacing a system is expensive not because the new technology is costly but because the organizational routines that have been embedded in the culture must be replaced.

The coordination that does occur between carriers today, happens primarily through personal networks that were built up over years in the job. This structure is operationally functional but not scalable.

4.2 Data Sensitivity and Trust

A topic that emerged consistently across the interviews is the opinions of carriers towards data sharing.
These opinions do surprisingly, not focus too much on the type of data shared but around the identity of the potential recipient and the relationship that the two actors have. The same operational information that one carrier shares freely and without reflection in one institutional context is guarded as commercially sensitive in another. This distinction matters and changes the platform adoption from a focus on which data needs to be protected to a question about who controls the data and under what conditions.

The most important entry point into this finding is the boundary carriers themselves draw between data categories. When asked what they consider shareable, interviewees consistently distinguish operational planning data from commercially sensitive data along a line that turns out to be less about information content and more about competitive consequences. The common theme is that operational data seems to be already known to most competitors and is hence not as secretive as other forms of data.

> *“Routes, timing, capacity, that I can share, that’s not secret. […] Pricing, customer names and specific volumes that’s another story. I would not be comfortable sharing that openly on a platform. Because then you basically give your competitors all the information they need to go after your customers.”* (Interview 5)

> *"It's like you already know all of our competitors' routes and what they are driving, which customer they have, you can see their trucks."* (Interview 6)

This classification of data being shareable in one instance, while categorically withheld in another captures the core finding about data sensitivity in this study. What changes between contexts is not the data but who receives it and what they can do with it. The mechanism that makes sensitive data genuinely sensitive is customer poaching. The fear that a competitor equipped with routing, volume or customer-location data, will use it to approach that client directly and then take them over.

> *"If you give the wrong information to the wrong person, suddenly they are calling your customer directly next week."* (Interview 4)

The rational response to this risk is not necessarily to withhold capacity in every case but to deliberatively develop a relationship where they see a benefit. In general, the benefit of sharing information is not only an optional add-on but a necessity. Potential gains range from access to return loads in unfamiliar markets to access to temporary capacity during peak periods. The latter is something that is crucial for carriers to offer, as the customer would otherwise approach their competitors directly. On the other hand, carriers will accept operational inefficiencies rather than competitive exposure. This logic is documented across multiple informants and reaches its sharpest formulation in a convergent statement from carriers in different regional markets:

> *"We'd rather drive half-empty than give the booking to someone who might steal the customer."* (Interview 8)

The protectiveness is not simply cultural conservatism. It reflects a historical adaptation to a market structure in which thin margins and high customer concentration make every customer valuable and especially the small size of carriers leads to this protectiveness. Understanding this adaptation is the prerequisite for understanding why DPs build upon anonymization and algorithmic optimization fail to attract participation from independent carriers. Carriers are not mistaken about the value of collaboration but that their individual cost-benefit estimation does not support participation with the current setups.

The trust dimension of this decision is equally important and resistant to resolution through platform design.

> *"I want to have met someone before I put my business in their hands."* (Interview 4)

> *"Trust takes years to build. You just can't pick up the phone and find a new one every day. […] Once the trust is broken, it's done. We don't use that partner again, as simple as that."* (Interview 5)

The under chapter 2.3.1 mentioned trust developments (knowledge-based trust vs. identification-based trust) both state that the formation of trust takes time and interaction and cannot be produced by a platform.

The trust problem that follows cannot be resolved at the technical level of platform design alone. Agahari, Ofe and de Reuver (2022) demonstrate that multi-party computation can enable parties to compute jointly over shared inputs while keeping

those inputs confidential. However, the concerns go further as the condition is not just technical capability but institutional credibility.

> *"I would need to know exactly what data goes where, who can see it, and what the benefit is for me. And I would want it run by a neutral party. Not a competitor and not a big platform that will use the data to compete against me. Maybe an industry organisation, or a research institute."* (Interview 4)

This is a precise and demanding governance specification, and currently no platform meets them. The interviews have shown that carriers are not opposed to data sharing in general but they want to know what happens with their data and who uses it.
The insight that the platform design is fundamentally a governance problem rather than a technical one is reinforced by examining the contexts in which data sharing does work.

This is especially imminent when looking at carriers that are part of an LBC. The structure of them makes carriers relate to their centralen as one of their own which makes data flows that would otherwise be unthinkable possible. The data is not less sensitive in general, but the architecture and structure of the LCB removes the need to treat the other actors as competitors. The centralen is co-owned by its members which share the operational data between each other and it is therefore not necessary to disclosure the data to a third party. A contribution to a shared administrative infrastructure that the carrier itself governs. Sharing with the centralen is, in a meaningful sense, sharing with oneself.

> *"Within [name of LBC], we have the same interest. If [name of LBC] wins, we all win. We have no reason to compete against each other on jobs. […] The trust comes from the framework, not from the individuals necessarily."* (Interview 4)

The difference between the marketplace environment is stark. Customer relationships, capacity, operating times and the costs can all be deducted from route patterns. A competing company with this information can easily determine the customers, how often they are served, which depots they are coming from, and on what terms. This is the knowledge spillover that has been mentioned by researchers in HCT literature. But what the interviews indicate is that the phenomenon is at a more granular level than what literature usually considers (Cruijssen, Cools and Dullaert, 2007; Lindawati *et al.*, 2015). Carriers aren't just protecting the strategic information of their business but the operational data, that over time accumulates into a clear overview of one's business.

That the adoption problem is laying in trust was also confirmed by one of the interviewed experts who has experience from working in digital solutions for the RFT sector.

> *"The technical stuff is, it's solvable. The hard part is, who trusts whom, who gets the upside, who bears the risk […]. A lot of the platforms that have tried and not succeeded in this space have underestimated that."* (Interview 2)

In the terms of the Actor-Network Theory: The Interessement phase where the process of convincing actors that participation aligns with their interests cannot advance when the governance structure does not exist. Carriers understand the theoretical benefits of data sharing but they lack an institutional assurance that those benefits will not be captured at their expense.

This finding extends and partially reframes the Privacy Calculus framework. Privacy calculus treats sensitivity as a property of data categories and the decision to share as a subjective cost-benefit assessment made by individual firms. What the interviews revealed is that sensitivity is not a property of data categories at all but a property of the data sharing arrangement. The same category of data produces an entirely different calculus depending on who receives it and what governance structure dictates their use of it. This has direct implications for platform design strategy. Platforms cannot resolve the sensitivity problem by offering extensive privacy guarantees at the technical level. The sensitivity itself is not ultimately about privacy in the individual sense but about the type of governance at the institutional level. The LBC's success in enabling extensive operational data sharing is not the result of a superior technical setup but the result of an organizational structure that aligns the interests of contributors and recipients.

The common element in successful cooperation models has been the established institutional structure. For carriers operating outside such structures, which are still the majority in Sweden, this institutional solution is simply not present, and the trust problem therefore remains structurally unaddressed.

4.3 Value of Tacit Knowledge

A reoccurrence in the interviews that stands in contrast to the idea of a data sharing platform is that the current coordination arrangements generate genuine operational value that digital platforms have not yet managed to replicate. This is a crucial analytical factor because it changes the terms of the adoption question. Rather than framing platform underutilization as conservatism or a failure to see the benefits from the

carriers, the interviews consistently point to something else. A rational assessment from a human with the experience and possibility to act upon changes. That is something carriers rely on and is absent in the current digital alternatives. The design-reality gap that this can be related to is not simply a product of implementation failure, but it reflects a shortfall of the platforms. These platforms do not offer the necessary skills that are important in the operationally challenging RFT sector.

The value, which is most mentioned, is the current handling of operational deviations that is done by humans. The issue is that these exceptions occur on a regular basis because the coordination of transports is usually not simply the observation of a plan, but the handling of exactly these exceptions. The deviations that occur range from failed pick-ups because of missing documents, over rescheduling of delivery times because of delays on the route to breakdowns of hardware and the need to find a replacement vehicle. When a platform is responsible for the route planning, they might be able to find the algorithmically perfect way, but they lack a mechanism to conduct negotiations with another human which dispatchers do as part of the daily operational planning.

> *"I see an advantage that our customer can call and speak directly with the person who is in charge of the truck and he can tell them: 'hey this truck is 10 minutes late. Is that okay?' When you have these extremely large players you can't contact them. You can send an email to a chatbot and maybe in two days you get a reply."* (Interview 10)
>
> *"There is no system support for that in most systems. You call the customer and say, 'Hey, can you help me out here? I will be late. Is that okay?'"* (Interview 3)

Together, these quotes describe the same phenomenon. One describes it as a competitive differentiator and one as an operational reality. The ability to resolve exceptions through direct human contact is a service quality feature that small and medium-sized carriers can offer precisely because their coordination is person-embedded rather than system-mediated and can also be seen as the reason to choose a potentially more expensive smaller carrier. This matters because it reframes the narrative around carrier scale which often claims that being small is a disadvantage of small carriers but as it turns out people are a valuable part of the business.

The tacit knowledge that an experienced dispatcher has however goes further than just handling exceptions. It includes knowledge about which loading bays at specific terminals are difficult to manoeuvre, which partner carriers can be trusted for a same-day emergency load, which roads get closed or clogged in specific weather conditions and which customers have informal timing tolerances that differ from their contractual

requirements. This knowledge is accumulated through years of experience, transmitted through observation and apprenticeship and stored entirely in persons rather than in any system. This is already starting to become an issue and companies are aware of it but fail to avoid it entirely.

> *"We're losing tacit knowledge faster than the digital tools are replacing it."* (Interview 4)

The organizational concern implied here is that as experienced dispatchers retire and coordination gradually shifts towards a more digital process and a collection of operational knowledge is not being captured or transferred. The problem is not merely organizational in the sense of workforce management, but it is also structural. The knowledge in question resists the kind of codification that would make it available to a digital system. As one expert who has worked directly on the integration of traditional carriers into a digital operations platform describes:

> *"The more traditional the carrier, the longer it [the integration] takes, because you're introducing digital ways of working into a part of their business that's been on paper or phones for thirty years. […] Some of those traditional carriers have beautiful operational discipline we've learned from but translating it into our system requires structured data and structuring data from workflows that were never structured is the core challenge of digital transformation everywhere." (Interview 12)*

This framing repositions the digitalization from a question of willingness to adopt to a question of knowledge translation capacity. The challenge is maybe not only that carriers refuse to change their way of working but instead it is that the change involves converting operational expertise into structured data which current TMS systems can't do. The gap between current carrier practice and digital platform functionality is even growing faster now:

> *"Technology is moving so fast right now, so you cannot expect a transportation company, even if they are willing to change […], to have deep knowledge of AI."* (Interview 3)

The data platforms require structured data inputs, while effective coordination in the current system is built on unstructured, expertise. Closing the gap requires a prior investment in making the tacit explicit which especially in a low margin sector is a substantial undertaking for the actors involved. Closing the design-reality gap in this

dimension requires a prior investment in making the tacit explicit, a costly and uncertain undertaking for which there is currently no clear institutional sponsor.

4.4 Institutional Enablers

This section evaluates the ability of existing institutional frameworks to resolve the tensions inherent in horizontal collaboration, using the Swedish Lastbilscentral (LBC) model as a primary case study. By examining how regional alliances like OnRoad try to mitigate competitive risks, the following analysis aims to demonstrate that while governance bridges can significantly reduce the design-reality gap, they are not a total solution. Structural barriers existing in Sweden define the practical limits of digital coordination in the RFT sector right now and illustrate why technology alone cannot fully optimize a fragmented market.

The LBC is best understood as a governance solution to the competitive conflict that prevents HC in the broader market. The co-ownership of the entity by the carriers has several structural consequences. As mentioned in chapter 4.2, it eliminates the exposure that makes data sharing dangerous in other contexts and solves the compensation problem that is commonly cited as a cause of horizontal collaboration failure (Frisk *et al.*, 2006; Aloui *et al.*, 2021).

> *"The lastbilscentraler are actually a pretty clever structure. A form of collaboration that has been working for 80, 90, 100 years. Gives small carriers access to customers and admin they could not access alone, while letting them keep ownership and independence."* (Interview 12)

> *"In a lastbilscentral the small [carrier] with one or two trucks, they need the centralen because otherwise they would never get customers or manage the administration."* (Interview 5)

This dependency of small carriers is precisely what aligns incentives between members. In practice, LBC carriers share operational data because the governance architecture has altered the meaning of the disclosure. However, not all carriers believe that an LBC is of value for them, especially the carriers that operate in an unique location where competition is not as strong and those that have been working with specific customers for a long time.

> *"Why should I pay six, seven, eight percent in commission to [LBC name] for them to send an invoice to the customer? I can do it myself [...] if you have been in this [industry] for a couple of years, like 10 or 15, then you*

can do it yourself." (Interview 9)

The OnRoad coalition extends this principle across regional boundaries and as a network of LBCs and carriers, they coordinate cross-regional transport throughout Sweden (*OnRoad - National Transport Network*, 2026). It achieves horizontal data sharing at national scale by combining the LBC governance architecture with geographic overlap between the members core markets. Interviewees that are a part of the collaboration distinguish OnRoad from other alternatives as follows:

> *"OnRoad is the opposite design philosophy, keep the relationships, add a digital layer on top. It works better socially because nobody feels disintermediated. The trade-off is the network is smaller and harder to scale beyond the members you already have."* (Interview 12)

> *"Yes, that's definitely a part of the success [regional non-overlap]. I also think one of the USPs of On-Road is that we are both national so we can reach the entirety of Sweden but we're also at the same time local because every region has its own owner and they know their own region very well."* (Interview 14)

This means that the design preserves relationships which helps to preserve the skills of individual partners making the system function. This design is a direct response to the fact that anonymous counterparties, algorithmic routing, and open access are precisely the architectural features that carriers reject.

However, even within these structures, the limits of the used simple digital platform become apparent.

> *"The most commonly used [planning tool] is T5 Barfors, but of course it only relies on human input, and the connections are not really in place. And especially then the communication also between the centralen and the different [carriers] they have, it still consists mostly of phone calls and mails."* (Interview 3)

Peet, Bharosa and Janssen (2025) argue that trust frameworks in multi-actor data-sharing settings must be actively maintained rather than assumed to persist once formally established. The persistence of human coordination within LBC structures suggests that even working frameworks cannot fully substitute the phone communication. The LBC model shows that horizontal data sharing at operational depth

is possible, but it does not show that the human coordination layer can be made superfluous by governance design alone.

Cost and information control can directly be related onto the two dimensions of the design-reality gap: technical and governance. Neither of them can be resolved in isolation which means that both must be addressed before full integration becomes rational for principal carriers managing complex subcontractor networks.
In the interviews, a third category of constraints, the structural barriers, came up. These barriers are rooted in the economic geography of freight demand.

> *"There is imbalance in the market all the time where there's more coming into Stockholm than coming out"* (Interview 11)

> *"The stats you see, 20%, 30% empty running, they include a lot of transport types where an empty return leg is literally the product. Gravel trucks, tippers, tankers. Those are structurally empty-return businesses, and no platform is going to change that."* (Interview 12)

One aspect that stands out is the geographic imbalance of demand leading to empty return trips. This occurs independently of the quality of the coordination process used. However, it is a problem that cannot be solved as it would mean that the entire industry needs to be relocated to balance the demand in those specific regions. Another thing that became apparent is the fact that empty running figures include a great variety of events which leads to skewed values.
The disaggregation has direct implications for platform scope and for the realistic assessment of achievable gains. While Carvalho and Sousa (2020) estimate that between 20 and 25% of truck trips in Europe are made completely empty, the interviews suggest that a substantial share of this figure reflects structural demand imbalances and vehicle-type constraints that platform-based matching cannot alter. The efficiency opportunity is real but specifically located in LTL and part-load freight corridors where matching failures driven by information fragmentation are the binding constraint.

A further structural constraint mentioned is the planning constraint in the industry.

> *"They tell [carrier name] we want 35 trucks. They are supposed to look like this. You have to arrive at our warehouse this time and then you load and then you come back at this time and so on and so on."* (Interview 11)

The planning constraint here is contractual. Carriers driving for major shippers lack the scheduling flexibility that opens the possibility for dynamic load matching or optimised planning making data-sharing tools irrelevant to them. Grote *et al.* (2023) find that even

in the public sector, barriers including service level agreements and liability concerns which result in a scenario where the achievable gains from shared-fleet logistics are erased.

Lastly, the total consolidation of the market on one platform, while raising anti-trust concerns, does not seem like a feasible solution. This means that there is also an upper structural limit to the adoption of a platform.

> *"What I don't think will happen is one dominant collaboration platform everyone uses. The industry is too fragmented, too regional, too attached to existing relationships. You might get regional leaders or sector leaders, but not the Uber of freight."* (Interview 12)

When considered at the same time, these structural insights highlight a mix of variables that affect the market simultaneously and not singular design-reality gap in Swedish road freight transportation. The gap encompasses a failure of governance and trust that can be resolved through a good institutional design, failure of knowledge translation that would require a considerable investment, and a structural mismatch that is partially unresolvable.
These dimensions are all correlated and a carrier that would be suitable for an adoption finds that certain factors make an adoption not beneficial. Closing the gap meaningfully requires engaging all three dimensions simultaneously rather than optimizing for any one while neglecting the others.

4.5 Summary

The empirical findings presented in this chapter describe the design-reality gap in the Swedish RFT sector from three directions, each corresponding primarily to one of the sub-research questions that structure the study.

Chapter 4.1, dealt with the current state of data exchange happening among carriers. The findings were a sector that is computerised but not digitalised. Coordination is primarily based phones and mails, and personal relationships are an important factor. The cause for this is the routines that carriers have built around themselves. Any data standardisation becomes a challenging task before each carrier is in a different stage of digitalisation.
The second direction described how carriers perceive and manage data-related risk. The key finding is that data sensitivity is depended on the context of data. The same operational information that is shared within an LBC is withheld on an open

marketplace. The interviews showed that the sensitivity depends strongly on the recipient and the conditions under which the data is shared.
A third finding from the interviews was that the value of human interaction still plays an important role in the industry. Carriers believe that this tacit knowledge cannot be replaced by a system yet. However, as especially smaller carriers view this tacit knowledge as a key part of their offering, the adoption of a DP is still slow.
The last key finding was about the structural constraints that limit what coordination can realistically be conducted on platforms. Geographic demand asymmetries, limitations depending on the type of transport and contractual inflexibility all contribute to the empty-running phenomenon and cannot easily be resolved.

Together, the three directions suggest that the design-reality gap in this sector is not the product of a single identifiable barrier but an interconnected phenomenon in which governance, knowledge translation and structural dimensions interact.

5 Discussion

The findings in the previous chapter reframe the underutilisation of data sharing platforms in the Swedish RFT sector as a connected phenomenon in which data sensitivity, governance and structural constraints interact. These three empirical dimensions can be combined into a single point. The design-reality gap in the Swedish RFT sector. It is not the product of a single implementation failure but a result of multiple connected issues. Resolving one while leaving the others in place does not shift the adoption picture in any meaningful direction instead, it simply displaces the binding constraint.

This discussion chapter connects the findings with the three theoretical frameworks introduced in chapter 2. The three subchapters of 5.1 correspond primarily to the three sub-research questions, while chapter 5.2 concludes the theoretical and practical implications that follow from the combined findings.

5.1 The Design-Reality Gap

5.1.1 Data Sensitivity as a Relational Property

The Privacy Calculus framework mentioned in the chapter 2.3.2 assumes a cost-benefit logic is pursued when data disclosure decisions are made. Their risk beliefs and confidence or enticement beliefs are weighed against each other. In relation to the organizational structure that applies to platforms within the RFT sector, some type of data, like customer details and pricing, are considered sensitive information and hence not openly shared. On the other hand, route and capacity information, are not treated as sensitive and are therefore shareable (Jussen, Schweihoff and Möller, 2023).

The findings from the empirical study done as part of this study, however, directly challenge this assumption. They claim that it is not the operational data that determines whether such data is freely shared in one institutional environment whilst being withhold in another, but it is the identity of the recipient. As illustrated in chapter 4.2., what kind of data is being shared depends on who is receiving it and a clear example of this can be seen when comparing a marketplace to the LBC context.

At first, this does appear to confirm the categorical assumption which the Privacy Calculus framework establishes. However, when asking the carriers about this, it

appears that the boundary of data sensitivity shifts with regard to the LBC setting. There information about route patterns, customer locations, and transaction volumes is often exchanged between the centralen and hence indirectly with other carriers. The data itself is not considered any less sensitive, but as part of an internal administrative function rather than an external disclosure.

Therefore, the same information can be both sensitive and not sensitive, depending on the receiving party and the governance structure involved. This insight changes the concept of data sensitivity from one that depends on the information into one that relies on the context of the information sharing and is hence a relational property. The reason why a route is considered sensitive is not the route itself but the possibility that a competitor can use it to approach the customer behind the route.
Due to the nature of the business, parts of the operation are visible to the competitors because the trucks drive on public roads. This information is not and also cannot be protected. What carriers are protecting is the aggregation of operational data into one connected picture that once handed to a competitor becomes valuable.

The implication for the Privacy Calculus model is that the benefits side, which can neutralise reversed a negative balance, cannot be manufactured by privacy guarantees alone. Anonymisation and Multi-Party Computation address privacy in the individual sense by protecting the confidentiality of the inputs, but this is not what carriers are protecting against (Agahari, Ofe and de Reuver, 2022). The concern is more about a competitive governance mechanism at the institutional level. A platform may technically encrypt every interaction but still leave the underlying risk exposure untouched, because the recipient remains a competitor. Given the nature of a competition, this leaked information would be used to approach clients and gain a benefit.

In addition, this reasoning is further strengthened by a cost asymmetry that is not adequately captured by the cost-benefit framing of the Privacy Calculus. The efficiency gains from sharing route or capacity data are relatively small on an individual level and only when distributed across the platform population and accumulated show clear benefits. The competitive exposure however is concentrated and individually borne. A single instance of customer poaching is enough to destroy years of building a trusting relationship. In a market where margins are small and fragmentation is high, even a small probability of such an event makes the individuals lean against participation. The underutilisation of digital platforms in this market is therefore not the outcome of behavioural conservatism or technological avoidance but an individual response to exposure. Importantly, this also reframes the gap as a collective problem rather than an individual behavioural one. Each carrier acts rationally given the other carriers

behaviour, but the collective outcome is the persistence of underutilisation. This is a reoccurring issue that no single actor can resolve on their own.

This extension of the Privacy Calculus has consequences for how platform formulate their design strategy. Platforms cannot reduce sensitivity by working on the data side alone, if sensitivity is relational. Instead, they have to work on the recipient side, meaning that governance conditions for the usage of data need to be specified. The LBC's success in enabling extensive operational data sharing is not the result of a great technical setup but the result of a governance structure that aligns the interests of contributors and recipients.

This insight is also the bridge to the next finding, where the question of governance as a precondition for platform participation is explored in more detail.

5.1.2 Governance as a Precondition

If data sensitivity is dependent on the context and the recipient, then the conditions under which sharing becomes acceptable is a question of governance. The empirical findings suggest that these governance conditions are not a design feature that platforms can add alongside their solutions. Instead, they must be in place before carriers will engage with a platform's value proposition at all.

The ANT, applied to inter-organisational data sharing by Jussen et al. (2024), describes the formation of a data-sharing network as a process of translation that proceeds through the four ANT phases mentioned in chapter 2.3.2. Applied to data platforms specific to the freight sector, the literature states that the platform itself is a step through which efficiency gains in horizontal collaboration can originate. The Problematisation has been articulated in the literature by Carvalho and Sousa (Carvalho and Sousa, 2020) and others who quantify empty running and underutilisation as systemic inefficiencies that data sharing could in principle address through optimised matching and load consolidation.

The empirical findings of this study suggest that the Problematisation phase has been largely successful. Carriers are aware of the inefficiencies that platforms aim to address and also the way they aim to address them. They acknowledge the logic that data sharing could in principle help to reduce empty kilometres and increase fleet utilisation at the same time. However, the Interessement phase, where the host of the platform must convince the other actors that participation aligns and benefits their interests, is where the translation process stalls. The interview findings indicate that this failure is not caused by carriers misunderstanding the value but by the lack of a governance structure that meets the conditions carriers see as preconditions for a participation.

These governance specifications are clear requirements and currently no platform meets them. The resistance from the carriers is therefore not a failure to recognise the potential but reflects the institutional constraints they face. The constraint lies at the level of governance and trust rather than at the level of technical capability. This finding shifts the explanation of platform underutilisation from a technical issue as framed in the literature toward an institutional problem. Trust in this sector is therefore better understood as an output of governance than as an input that platforms can supply through their design and structure.

Bringing up the LBC model as a comparison here is analytically central to this argument and deserves to be looked at in some detail. Within an LBC, the architecture has already pre-resolved the competitive exposure that makes the Interessement phase so hard to overcome elsewhere. Its ownership model transforms the identity of the other actors from "competitors" to "co-owners". Sharing operational data is almost as if you share the data with oneself. The entity itself benefits from the data and hence also the participating carriers. The translation has already been completed at the institutional level before any digital platform was introduced within LBC structures. This explains why data sharing at an operational level is achievable within LBCs and consistently fails to function outside these structures. The LBC approach to the data sharing problem through governance is something that digital platforms have not managed to do despite all their technological advances. The lesson that can be learned from this comparison is that the underutilisation of data sharing platforms does not have much to do with the lack of proper platform design, but rather with its composition problems. The relevant question is hence not which features the platform should have, but which governance structure the platform should be embedded in.

This finding extends the concept of platform translation in ANT theory. According to ANT practices, the Interessement process includes three elements, demonstration, prototype and economics arguments. They play an important role, with the underlying assumption that the host will manage to get the other actors convinced to participate in the Interessement process. However, in case of markets characterized by thin margins and a high fragmentation, demonstration of efficiency benefits will not be able to offset the fear of potential losses due to participation in the process by the carriers. According to Dinev & Hart (2006) argue, trust can override risk as long as there is sufficient basis for it. However, it is evident from the findings of this research that the existing platforms do not enjoy this credibility level yet. The governance architecture is therefore not a design feature but a precondition for the Interessement phase to be entered successfully at all.

The implication for the broader question of whether digital platforms are even the right instrument for HC in the RFT sector is not an easy question to answer. The empirical material does not support the claim that DPs cannot work in this sector but throughout the interviews there was also no clear evidence of a DP that is used. The OnRoad coalition demonstrates that horizontal data sharing can scale across regional boundaries when the governance architecture is adapted and a digital layer is added on top of an existing institutional structure (*OnRoad - National Transport Network*, 2026).

What the interviews suggest is that platforms, as currently designed in the open-marketplace tradition (S3 in the typology of Pan *et al.*, 2019), are not yet ready for the RFT sector. Their choice to value algorithmic matching and anonymisation over institutional trust leaves the problem they actually have unresolved. The emphasis on privacy solutions reflects the broader assumption in the DP literature that the obstacles to inter-organisational data sharing are primarily technical (Möller *et al.*, 2024; Pampus *et al.*, 2025). The findings here suggest that this emphasis is at least partially invalid when transferred from environments with existing data exchanges to environments where the recipients are direct competitors.

Digital platforms have a meaningful role to play if the governance architecture in which they are embedded resolves the competitive exposure problem first. Without this prior step, the platform's Interessement phase fails because the institutional preconditions for participation have not been met and not because the technology is inadequate. The question for future platform design in the sector is therefore where to embed them, and a potential solution could be an ownership model based on the LBC model.

5.1.3 Structural Limitations of Platforms

This last past deals with the differences in between the current state of planning and data exchange in the RFT sector and the deliverables of the platforms. This gap can be related to the design-reality gap. It was developed by Masiero (2016) and applied to the Swedish RFT context by Larsson, Linåker and Runeson (2024) and notes that platforms are designed under a set of assumptions that do not consistently reflect the operational environment in which the platforms are deployed. The interviews conducted for this study suggest that the gap in this dimension is wider and more structural than the existing literature typically acknowledges. Three key structural points emerged from the material and point to limits on what platforms, can achieve in the current RFT sector.

The first point is that the digital transformation in this sector is often a knowledge translation problem. The current coordination process among carriers creates a value which cannot be reproduced by the currently existing digital technologies. In particular, it concerns the coordination of operational exceptions using human connections and

storing implicit knowledge related to loading areas, partners, and customer preferences which are different from those agreed upon contractually.

This human layer of coordination is not just an issue which is yet to be digitalised but rather a key component of the services provided by small and medium-sized carriers. To bridge the design-reality gap, not only the introduction of new technologies is needed but also the initial coding of unstructured operational knowledge is required.

However, the required transition of the human knowledge requires significant investments both in infrastructural development and human capital. Some loss of flexibility that inevitably comes along such a transition is one thing that the existing industry cannot tolerate. A counter-intuitive consequence of the situation is that the scale not only does not promote the digitalisation but may even deepen the problems. Larger carriers own more legacy technological tools and systems and therefore might have even stronger path dependencies. This means that the process of changing the old procedures takes more time and financial effort than the smaller firms might need to invest. It results in the fact that the largest potential users are precisely those who find it difficult to meet all the prerequisites of digital transformation, such as standardisation or an integration of systems. Thus, the problem of adopting the technological platform cannot be regarded exclusively as the problem of small carriers, but for carriers of all sizes. Particularly, those that have worked for many years, adapting none of their procedures and relying on infrastructure that is not designed for an integration.

The second aspect relates to the way of interpreting the statistics behind the empty-running figures that serve as the reason for numerous studies on collaboration along the HC literature. The frequently quoted estimate of 20-25% of the total number of trucks in the EU operating empty (Carvalho and Sousa, 2020; Eurostat, 2024) includes all types of services, despite their differences that makes such a collaboration impossible. As one participant notes, a large portion of these percentages can be explained by asymmetries in geographic demand, constraints in segments such as tankers and gravel trucks where empty running is unavoidable and contractual inflexibility imposed by major shippers. These factors leave carriers no room for dynamic load matching and pose the question of how much of the empty runs can actually be reduced.

This has direct implications for the scope of a platform and for the realistic assessment of gains and benefits. The opportunity that platforms can realistically address lies in the LTL industry where information fragmentation and load matching is the binding constraint. The framing of the inefficiencies as a single number asks for a response of platform consolidation where one platform is the solution that fits all the needs, but the inefficiencies do not actually support this. Platforms that target the specific areas where

failure is caused by a lack of information seem like a better way to create a business model. Especially when comparing it to the empty runs caused by a structural demand pattern, the lack of information is an issue where collaboration would actually generate value.

The third structural point revolves around an upper limit on platform consolidation that exists independently of the previously mentioned points. A very fragmented and regional market that is very embedded in bilateral relationships creates a challenging scenario for one dominant platform to emerge. In addition, such a consolidation would raise antitrust concerns and would have to be carefully monitored to prevent advantages for certain actors with a lot of power.

This observation is consistent with the OnRoad collaboration mode. Here the digital layer is deliberately fit to the size of the institutional cooperation rather than aspiring to cover the full market. It also reinforces why governance architecture is the relevant design variable for platform development in the sector. A platform that aspires to a full market integration faces challenges from both the market fragmentation and the different requirements from different types of transport. Neither of these can be engineered away through better matching algorithms or more attractive pricing structures.

Taken together, the three structural points reframe the design-reality gap as a phenomenon with several dimensions that act independent from each other. The governance dimension developed in chapter 5.1.2 can in principle be addressed through different institutional designs. The knowledge translation dimension can be addressed but requires substantial prior investment. But the structural dimension is partly outside the reach of platform design altogether or at least limits the overall scalability which would in return also limit its benefits. Closing the gap meaningfully therefore requires engaging in all three dimensions simultaneously. A platform strategy that optimises along one dimension while neglecting the others will, as the Privacy Calculus Theory in chapter 5.1.1 already suggests, worsen the cost-benefit calculation for the carriers rather than improve it. Interventions acting on one dimension predictably fail to shift adoption rates because the constraint simply shifts to whichever dimension was not appropriately addressed.

The findings could also be extended beyond the Swedish RFT sector. The combination of thin margins and a high fragmentation is not unique to the RFT sector. Instead, it has been documented in adjacent settings such as the Norwegian fishing fleet (Sønvisen et al., 2025) and public sector logistics (Grote et al., 2023), where independent actors compete on operationally interdependent activities. The barriers documented in the present study therefore appear to be applicable across other industries operating under

similar conditions. The contribution of this study could therefore be seen as broader than a sector-specific finding on the Swedish RFT sector and closer to a contribution to the theory of platform adoption failure under certain conditions.

5.2 Contributions and Implications

5.2.1 Theoretical Contribution

This study looked at different theoretical lenses that have so far been treated separately in the data sharing literature. The primary theoretical contribution of this study lies in extending the Privacy Calculus framework as it has been applied to inter-organisational data sharing. The ANT and the design-reality gap serve as complementary theories to contextualise the main finding, but the study does not extent those frameworks. Instead, applying the theories together to the fragmented and competitive RFT market can help to explain the platform adoption failure.

The first contribution complements the Privacy Calculus framework as introduced by Dinev and Hart (2006) and applied to inter-organisational data sharing by Jussen, Schweihoff and Möller (2023). The framework treats data sensitivity as a property of data categories and the decision, whether to share data, as a cost-benefit assessment. The empirical material in this study show that this assumption requires some clarification. The same operational data is treated as freely shareable within an LBC and as commercially sensitive in an open marketplace. The reason for some data to be classified as sensitive is the possibility that a competitor can use it to approach the customer behind the route. Therefore, sensitivity should be treated as a relational property of the data sharing arrangement rather than a property of the data itself.
The confidence and enticement side of the Privacy Calculus theory is not generated by privacy guarantees at the technical level but by the governance architecture. Especially in a competitive setting, the governance architecture defines who the recipient is and what the recipient is allowed to do with the data.
This finding extends the Privacy Calculus framework. It was already shown that trust in the recipient can override risk perceptions. What this study adds is a structural account of how that trust is produced. In the LBC context, this is achieved through co-ownership and pre-established governance rules. These findings suggest that future work should account for the identity of the recipient and the context in which the data is shared. If these factors are neglected an important measure may be overlooked.

The second finding concerns the Interessment phase of the ANT when being applied to data-sharing contexts. The literature describes the Interessement phase as the stage

where the host must convince the other actors that a participation benefits their interests which is often achieved through demos, prototypes and economic arguments. The findings here show that in the RFT market with thin margins and high customer concentration, these methods have not been able to convince carriers to adapt a platform. Carriers understand the value a platform could deliver in regard to empty running and increasing utilization. What carriers are missing, is an institutional structure that makes it viable to participate in the Interessement phase. The standard ANT therefore needs to be complemented with an additional precondition that states that governance system must be restructured for the Interessement phase to be successful.

The third contribution concerns the single dimension framing that the design-reality gap framework uses. The existing literature applies the framework to single-dimension issues such as data standardization or partner interchangeability. The findings here show that in a highly fragmented market, the gap is often connected and acts simultaneously across all dimensions. The most important implication of this connected framing is that the dimensions interact with each other rather than simply complementing each other. Resolving one dimension while leaving the others unaddressed is not neutral and could even worsen the carriers' cost-benefit calculation. A platform that resolves the governance question through co-ownership but ignores the knowledge translation question forces carriers to make a digital transition for which they are not yet equipped. While a platform that resolves the knowledge translation question but ignores a fitting governance structure exposes the carriers to data exposure risks.
The design-reality gap therefore needs to observe and consider all dimensions at the same time to achieve adoption. Partial fixes that have dominated the platform landscape so far are ineffective and have failed to achieve a breakthrough.

Taken together, these three contributions have the potential to be applicable beyond the Swedish RFT context, but the extrapolation of results must be done carefully. The main findings of the study, namely the relational nature of data sensitivity, governance as a precondition and the combined character of the design reality gap, are derived from structural conditions that are not unique to Swedish road freight. Fragmented markets with thin margins that face similar issues exist and possible solutions have already been briefly mentioned in chapter 2.4.3. The existence of similar barriers in other sectors suggests that the underlying mechanism could occur in fragmented markets operating under exposure risk rather than being specific to road freight.

The transferability of the results needs to be further explored because existing structures like the LBC structure are unique to the market in the Nordics. The broader EU RFT

sector does not generally exhibit an equivalent of the LBC even though they operate under similar structural conditions. This might lead to governance solutions that resolve the Interessement problem in Sweden failing in other parts of the EU because the market changes drastically in between countries.

The contribution of this study is therefore best understood as a conditional one. Data sensitivity and governance preconditions were identified as the key contributors to a platform adoption failure of competitive exposure. The solution, however, is presented as a Swedish demonstration of one possible form of governance pre-resolution rather than as a universal blueprint. The task of identifying the institutional form most appropriate to other national markets and other sectors is left to future research and is taken up explicitly in chapter 5.2.4.

5.2.2 Practical Implication

The practical implications of this study address three key stakeholder groups that can derive concrete priorities from the findings.

The first stakeholder are platform designers. The most direct implication is a reorientation of the design priority from technical privacy solutions to institutional governance design. The current generation of data sharing platforms in the RFT sector emphasises technical privacy through anonymisation and encryption. These technologies have value, but they address a problem that the findings show is not the binding constraint. Carriers are protecting themselves against the institutional risk that a competitor will use any received information to approach their customers. The implication is that resources currently allocated to privacy technologies will not deliver a substantial increase in usage. A redirection of design effort towards governance models that align the incentives of contributors and recipients seems to align better with the interests of the involved parties.

The second stakeholder are the carriers themselves, who are at the centre of this study. The findings have direct consequences for how carriers should evaluate the platforms and digital initiatives that are increasingly being presented to them.
Firstly, the interviews show that carriers already understand the theoretical benefits of collaboration but protecting their data is still a reasonable behaviour. The decision to participate in a platform should be made on the governance structure of that platform rather than on its technical features alone.
In addition, the findings indicate that dispatchers with tacit knowledge and exception handling skills cannot be replaced by digital tools yet. Carriers that are planning a digital transformation should therefore plan investments not only in the platform itself but in the prior work of converting tacit operational expertise into the structured inputs that a

platform requires. Without this step, the platform potentially performs worse than the human system it replaces. Carriers should also recognise that the human coordination layer in itself is a competitive advantage and that platform adoption does not automatically remove this asset.

The third stakeholders are policy makers. An implication for policy makers concerns the contribution that data sharing platforms can make to the carbon reduction of the transport sector. Currently, the transition to zero-emission vehicles is the main driver to reduce emissions. However, the capital weakness of the carriers limits the rate at which they can replace their diesel fleet with alternatives as current vehicles are more expensive than their counterparts.
The findings of this study suggest that a more immediate solution could be HC. Data sharing at an operational level appears to be achievable in the RFT sector when the recipient is aligned with the contributor. The LBC structure has done this successfully in Sweden and could be viewed as an exemplary case. In this context, collaboration is perceived as contributing to a collectively governed system. Building on this foundation, the LBC model could also be further developed to allow for a more structured data sharing between carriers. Integrating a shared data layer into the centralen's existing planning function could for example allow empty-runs to be spotted within the network. Since the competitive exposure problem is already resolved due to the governance structure, the remaining barrier is technical. Standardised data inputs can be used for an optimisation of the truck's utilisation. This would represent a next step from computerisation to digitalisation within an already trusted institutional framework.

Policy makers seeking to encourage data sharing could therefore consider whether existing legal and regulatory frameworks adequately support the introduction of cooperative governance structures and if not, what steps can be taken to facilitate it. A policy intervention that improves the conditions under which carriers can collaborate is not only an intervention but could also lead to reduced emissions due to efficiency gains. Improvements in the overall efficiency, represent a quick solution to reduce the emissions from the sector and one that does not require big upfront investments. The implication for policy is that improving the conditions for carrier collaboration and continuing the fleet electrification can happen in parallel.

5.2.3 Limitations

The findings presented above are subject to several limitations that should be considered when interpreting them. The limitations presented in the following section complement the constraints listed in chapter 1.4 and focus on the empirical boundaries that became apparent throughout the analysis itself.

The study with fourteen semi-structured interviews provides the depth necessary to identify mechanisms and relational patterns but it does not support statistical generalisation of the results. The findings indicate a structure in the market but do not deliver a statistically representative result. The interpretation of the material is, even though this was mitigated as much as possible, influenced by the decisions of the researcher. In addition, the interview sample itself might be a limitation. Carriers that were willing to discuss digitalisation topics in a research interview are potentially more likely to be more reflective about these topics than the average carrier in the sector. The sample is therefore likely to be skew towards carriers that already deal with digital solutions and aim to make their business more efficient. This is further strengthened by the fact, that the interviewed carriers are relatively large and therefore most of the market (small carriers) is not included in the analysis.
This has consequences for how the findings should be read. Conclusions about the willingness of carriers to restructure and to consider new platforms likely reflects the upper end of the openness distribution in the sector rather than its average. A larger and more diverse sample would be needed to assess how the views identified here change when a less reflected part of the carrier population would be included.

The focus on carriers and on a smaller number of industry experts means that shippers and other stakeholder were excluded from the data collection. This was a deliberate choice given that the research questions target the carriers, but it means that the findings reflect the perspective of only one stakeholder group. Shippers may have different perspective on the implementation of data sharing platforms because it could weaken their influence on the carriers. The conclusions therefore characterise the carrier side rationality but not the system wide adoption dynamics that result from the interaction of all stakeholder groups.

Another limitation concerns the role of the LBC model. It serves as the primary demonstration that a governance platform design is achievable, its mechanisms work precisely because the model is limited. LBCs members do not compete with each other for the same shipper contracts within the structure, which an open platform would fail to replicate. This shows the limited applicability of the LBC model on the whole market. In addition, since the LBC is the only structure examined in depth because it is the dominant cooperative form in the Swedish market, the study cannot rule out that its success in enabling cooperation is partly because of other factors. A second and related limitation is that even within LBC structures, the digitalization gap has not been closed. LBC members share operational data with their centralen, but coordination is not structured around digital data. The LBC therefore demonstrates that one dimension of the design-reality gap can be resolved, but it does not demonstrate that all three

dimensions can be resolved simultaneously, and the interactions between them remain empirically unexplored.

The last limitation is the geographic scope. The material is centred on the Swedish RFT sector which with the presence of the LBC has a setup that is not equally available in the EU. The argument that the LBC solves the exposure problem relies on the existence of these structures which have not been further studied as part of this study.

5.2.4 Future Work

Several research directions follow from the findings and the limitations outlined above that are mentioned in the following section.

The first potential future work deals with the empirical extension of the framework to other sectors. The study suggest that sensitivity is often relational and that a governance structure needs to be established before an implementation. These topics might be applicable to fragmented markets operating under competitive exposure rather than being specific to road freight. Replicating the analysis in other markets would help to establish the empirical boundaries of the contribution. Case studies between markets with and without functioning governance structures could also be extremely informative as they would help to understand the contribution of governance forms versus the influence of other factors.

The governance finding raises a question at the policy level of the EU. While the EU Data Act creates the legal right to data access, the actual implementation of the European Mobility Data Space faces some challenges. The insight from this study, that governance structure is a precondition rather than a design feature, could be relevant to these implementation decisions. Future research could examine how different governance models affect the adoption in fragmented markets which would allow the theoretical contribution of this study to be applied concretely.

The carrier side that has been developed in this study has revealed some interesting behaviours in the RFT market. The shipper perspective, however, is less well documented in this context. This leaves the question open whether the findings can be generalized into a system-wide dynamic or if the findings are counteracted by competing incentives from the shippers. A particularly interesting question is whether shippers with big market power could even encourage data sharing among carriers and if such an intervention would produce a genuine collaborative behaviour or only formal compliance.

A last potential topic of future research is the quantification of the results. The current findings rely on qualitative interview data and are therefore susceptible to the

interpretive behaviour mentioned in chapter 5.2.3. A survey that measures how different recipients and governance structure affect the willingness to share data could allow the findings to be tested.

This chapter has shown that the design-reality gap in the Swedish RFT sector is a phenomenon with distinct dimensions that interact in different ways making partial fixes ineffective. Each of these findings extends the theoretical frameworks applied in chapter 2. The Privacy Calculus, the ANT and the design-reality gap theory each capture part of the picture but require extension to account for the competitive and institutional specificity of fragmented markets. Taken together, the contributions suggest that the path forward for data-sharing in this sector runs through institutional design before platform design. In addition, the efficiency gains from collaboration remain depended on governance conditions that have not yet been established.

6 Conclusion

This study had the goal of explaining the persistent gap between the theoretical potential of data-sharing platforms for HC in the RFT sector and their practical adoption among Swedish carriers. Fourteen semi-structured interviews with carriers and industry experts were conducted as part of this study. The central finding is that this gap is not the result of missing technological capabilities. It is the product of structural conditions that carriers experience in a fragmented market.

The interviews described a sector that is computerised but not yet digitalised. Coordination still relies on phone calls and mails, and personal relationships dominate the daily operations which no system currently can replicate. Platforms that assume that the existing data is structured in any way face a market that is not yet ready. At the same time, carriers are not resistant to change but they protect their data as a response to the competitive environment they operate in. The sensitivity of data in this sector is not a property of the data itself, but of who receives it. A fitting governance structure is therefore a precondition for a wide adoption and not a feature that can be added later. The Swedish LBC model demonstrates that this can be achieved. Its ownership structure removes the barrier to sharing but they also have not been able to digitalise the operations.

Taken together, these findings explain the persistence of the adoption gap. Each issue is solvable on its own with the current tools. But because they interact, resolving one while leaving the others untouched does not resolve the adoption difficulties in any meaningful way. This study contributes to the existing literature by reframing sensitivity as a relational rather than a categorical feature, stating governance as a necessity for the ANT translation process, and showing that the design-reality gap in this sector is an interconnected phenomenon in which single-point interventions predictably fail. The most important boundary of these findings is their grounding in the Swedish context. In Sweden, the LBC structure exists as an established institutional form which makes the transferability to other markets difficult. However, it does establish the structural conditions that a possible solution would need to fulfil.

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Appendix

Declaration of the use of AI

This following section depicts to where and to what extent Artificial Intelligence has been used.

Used Program	**What was it used for**
ResearchRabbit (researchrabbit.ai)	The initial literature was uploaded and ResearchRabbit is able to generate a connected graph of the citations. This was used to find articles that have not been included in the literature review before but were cited by multiple articles.
Gemini (gemini.google.com)	Gemini was used to conduct an initial deep research about the topic to give the author a general idea. In addition, it was used to formulate various research strings that used synonyms to find as many relevant articles as possible. Lastly, Gemini was used to generate an initial transcription of the Interviews.
Claude (claude.ai)	Claude was primarily used towards the end to check the spelling and coherence of the report. Complicated sentence structures were simplified and the transitions between different parts was improved with the help of Claude. Lastly, Claude also helped to flag repetitions that occurred.

While AI offers a lot of potential and can help to simplify tasks, the use of it and its results need to be carefully evaluated. AIs often make up results that do not reflect the reality and especially when dealing with academic references seem to hallucinate.
To prevent this from negatively affecting this report, AI was only used to improve already existing paragraphs. In addition, if an output of an AI had some application in the final report, the output was cross-checked with existing sources before being incorporated into the text. The transcriptions of the interviews were compared in full length against the recordings and corrected if necessary. No AI was used to generate interpretations or conclusions based on the empirical material.

I hereby declare that the above information is accurate and that I have disclosed all AI-generated content that directly influenced or contributed to my master thesis. I understand that failure to disclose AI tool usage accurately may be considered a violation of academic integrity policies.

Student 1 name: Ferdinand Axel Blase

This submission represents my own original work. Any use of AI tools has been appropriately disclosed, detailing their purpose, extent of contribution, and adherence to ethical guidelines.

Interview Guidelines

This section shows the interview guidelines that have been used for the semi-structured interviews. The guidelines were used as orientation points and before each interview the questions were adapted to better fit the interviewee. Throughout the interview the focus was adapted to reflect the interviewees main area of expertise and questions that were less relevant to them were excluded. In general, the questions have changed throughout the interview process as new topics emerged and for this reason the interview guidelines have adapted correspondingly.

Interview Guideline Carrier
Introduction
• Short Intro and description of the purpose of the study • Aim of the interview: Understand how your company is involved in sharing data with other actors and which factors influence the willingness to (not) share data. • Time frame: 45 to 75 minutes. • Confidentiality: If permission is granted, the interview will be recorded for subsequent transcription and analysis. All content is treated confidentially, and transcriptions are anonymised. You have the right to withdraw at any time and the recording will be deleted after the transcription to ensure privacy.
Background
• Could you please shortly describe your company?   ○ Size (number of vehicles, employees)   ○ Other smaller carriers as sub-contractors   ○ Types of freight transport (heavy, consumables, moving)   ○ Domestic/ international   ○ Other modes of transport (maritime, rail, etc.) • What is your current role and what are your responsibilities in the company? • How long have you been working in this industry?

- What kind of transports are you doing?
    - Box / Side-curtain, Flatbed, Tanker, Refrigerated
    - FTL/LTL
- How digitalised would you say your company is?

Current setup

- What does your typical workday look like?
- How does your planning work and do you use software in this process?
    - How do you get data about drives (internal system/ via external software)?
        - Manual vs. automated
    - Do you use several systems for data collection (integration into their system/ Fleetboard)
        - If yes, what kind of data is collected via the system
        - Manual vs. automated
    - What do you do with it?
        - Manual vs. automated
    - Do you use any tools to optimize your delivery/ planning?
- What information do drivers receive for their drive and how do they get this?
    - Are their any feedback loops/ reporting of issues by drivers
- What is your utilisation rate?
    - Are there ways to improve it and if so, which ones?
    - To what extent do you see empty runs as a structural problem or something that can be resolved?

Horizontal Collaboration / Sharing

- Are you part of a collaborative structure (LBC,...) and if yes what benefits do you get there?
- Are there collaborations with external partners (HC) (carriers, freight forwarders)
    - To sell additional capacity

- What information do you share with them and is there information you hold back?
    - Distinguish between different types of data?
    - Delivery time, locations, capacity, status, type of goods, customers (rank them)
    - How is the information shared (mail, message, digitally)
    - Willingness to share data when aggregated?
- Are there benefits you expect from collaborating?
    - Increased demand, possibility to bid on bigger tenders, increased efficiency?
- Are there challenges when collaborating?
    - What do you think are the main obstacles/ barriers for sharing data?
    - Do you think that sharing data could foster poaching?
    - How do you think the costs should be shared?
- Trust:
    - Do you see trust as a decisive factor?
    - How did you build trust with them?
    - Would you register to a website where capacity can be bought or sold but you have to upload all your delivery data? If not, why?

Outlook

- In which process do you see the biggest need to become digitalized and what would be the additional value?
- What benefits and risks do you see when sharing information?
- What would make a collaboration easier for you (privacy wise)?
    - Via governmental authority/institution, private company
    - centralised/ decentralised system
    - Legislation (in case of conflict)

Ending

- Do you see any other factors, besides those previously mentioned, that influence the decision process for sharing data? Did we forget something you'd like to say?
- Do you know of any other experts we could interview on this topic, and can you help get in touch with them?
    - Potentially partners of yours

Interview Guideline Experts
Introduction
• Short Intro and description of the purpose of the study • Aim of the interview: Understand how data sharing in data ecosystems works and which factors influence the willingness and decision making to (not) share data. • Time frame: 45 to 75 minutes. • Confidentiality: If permission is granted, the interview will be recorded for subsequent transcription and analysis. All content is treated confidentially, and transcriptions are anonymised. Have the right to withdraw at any time and recording will be deleted after the transcription to ensure privacy.
Background
• What is your current role and what are your responsibilities?   ○ Main area of expertise • How long have you been working in this industry? • Have you in the past or are you currently involved in research projects, initiatives, or industry collaborations related to: ○ data sharing   ○ User adoption   ○ logistics platforms   ○ digital ecosystems

○ Carrier collaboration
Current State
• How would you describe the overall digital maturity of data exchange in the logistics sector? ○ Do smaller carriers use planning software? • How significant is the issue of incompatibility of systems to data sharing in practice? • What are the main drivers that encourage logistics companies to share data? • What are the most common barriers that prevent companies from sharing data? ○ Literature states a design-reality gap (difference between platforms and their rate of adoption) • Do you think that tacit knowledge is an issue for the industry?
Trust / Collaboration
• In your opinion, what role does trust play in data sharing between logistics companies? • What mechanisms help create trust in data-sharing environments? • What role do cooperative structures play in enabling data sharing? • Is there a preferred data sharing system in the industry? • Do you think logistics companies are ready for large-scale data ecosystems?
Outlook
• Do you expect more centralized platforms, decentralized systems, or hybrid models? • What changes would be necessary to increase companies' willingness to share data? • From your perspective, what is currently missing in research or industry discussions about logistics data sharing?

- Are there important aspects of this topic that we did not cover?
- Do you know other experts, researchers, or practitioners who would be interesting to interview?

Interview Codes

Selected examples of each first-order concepts are described in the following part.

Dimension 1: The Issue of Data Exposure

Data Shareability as Relational and Contextual

Sharing of Operational data

“We share as much as we can because it raises questions and manual labour from our carrier partners if we don't. But we are held back by different TMS systems from ours and our partner, so everything isn’t transmitted.
- Interview 10

Withholding of Pricing and customer data

"Whereas if you start talking about route data, customer information, pricing, that is completely different. Companies are very protective of that kind of information."
- Interview 13

Willingness to share tied to relationship and competitiveness between actors

“Our partners and competitors are often the same. We small companies have to find a flexible relationship where we help each other with the goods and transportations.”
- Interview 10

Trust as a Structural Precondition

Personal trust as a prerequisite for sharing

“We are trying to choose maybe not the cheapest subcontractors or partners but those that we see a long-term relationship with. Mainly so we can prepare and set up connectors to then also share relevant drive information.”
- Interview 10

Relational continuity builds and signals trust

“I think most of our subcontractors they’ve been with us for a very long time. I started the first time at [carrier name] in 2003 and [...] came back in 2022. And I would say I recognize most of the subcontractors.”
- Interview 11

Trust breach and the impact on the relationship

“You can’t put it in writing right so it has to be some kind of mutual agreement not with words but in trust maybe.”
- Interview 11

Competitive Exposure and Customer Poaching as a dominant risk

Customer poaching in connection to data control

"It's also very delicate in who you choose to collaborate with and who you choose to go to first when you're selling a load. So yeah, it's definitely something you have to be aware of."

- Interview 14

Withholding data as a competitive defence strategy

“Of course it happens and it’s not much you can do about it too because of the competition laws. [...] We do try to protect us against that [customer poaching] and don't tell them everything when we want to sell additional capacity”

- Interview 11

Reputation as governance systems in regional markets

“Carriers don’t see themselves as interchangeable nodes in a network. They see themselves as businesses with customers, relationships, reputation. Handing over a shipment to a platform that routes it to whoever has capacity feels like losing the relationship.”

- Interview 12

Platform Scepticism and the Value Proposition Gap

Spot markets incompatible with high quality product

“It’s [spot market] not B solution, it’s not a C solution, maybe D or E solution to use it.”

- Interview 10

Gap between platform promises and carrier benefits

“The problem is not the platform, the problem is trust and quality assurance. If a platform could somehow guarantee that the carrier I am handing a job to is reliable, insured, HVO-capable, then I might use it.”

- Interview 4

Demand for relationship-preserving platforms

“Digital tools should support the relationship, not replace it. [...] If I could say: I share my capacity with partner A but not partner B, that would solve a lot of trust issues.”

- Interview 5

Dimension 2: Trust-Governance Interdependency

Governance Structure as a Trust Mechanism

LBC and åkericentral ownership model as a pre-resolved competitive conflict

"Trust is held somewhere in the institution. And the LBC is a very specific kind of institution where the hauliers are partial owners."
- Interview 2

Franchise model as a model to harmonize competing interests
“Within [name of LBC], we have the same interest. If [name of LBC] wins, we all win. We have no reason to compete against each other on jobs. [...] The trust comes from the framework, not from the individuals necessarily.”
- Interview 4

Demand for neutral platforms in order to participate
“I would need to know exactly what data goes where, who can see it, and what the benefit is for me. And I would want it run by a neutral party. Not a competitor and not a big platform that will use the data to compete against me. Maybe an industry organisation, or a research institute.”
- Interview 4

ICT Fragmentation and Path Dependencies from the past

TMS fragmentation blocks cross-organizational data flow
“We are held back by different TMS systems from ours and our partner, so everything isn’t transmitted. So when we have some extra info it could be like crane delivery or something outside the normal distribution tracks, we have to send an email.”
- Interview 10

Analog coordination habits embedded as organizational default
“The most commonly used [planning tool] is T5 Barfors, but of course it only relies on human input and the connections are not really in place. And especially then the communication also between the centralen and the different åkerier they have, it still consists mostly of phone calls and mails.”
- Interview 3

Path dependency of investments as a constraint on digitalisation
“The digital maturity in the logistics sector is stuck in 1999. Basically. They have computerized the system, but they have not gone digital. The processes are still in many segments based on the analogue precursors to them that were developed before 1999.”
- Interview 3

Dimension 3: Design-Reality Gap as Rational Equilibrium

Tacit Knowledge and Dispatcher Coordination

Dispatcher tacit knowledge not yet codifiable

“These systems are still very much dependent on humans. And when it depends on humans, you have linear scaling laws. If you want to double the volume, you have to double the amount of people in the system to handle these types of things.”

- Interview 3

Phone-based coordination is important but unscalable

“I see an advantage that our customer can call and speak directly with the person who is in charge of the truck and he can tell them hey this truck is 10 minutes late. Is that okay? When you have these extremely large players you can’t contact them. You can send an email to a chatbot and maybe in two days you get a reply.”

- Interview 10

Digital transformation as a translation challenge

“The more traditional the carrier, the longer it [integration takes], because you’re introducing digital ways of working into a part of their business that’s been on paper or phones for thirty years. And the older way isn’t necessarily worse. Some of those traditional carriers have beautiful operational discipline we’ve learned from but translating it into our system requires structured data, and structuring data from workflows that were never structured is the core challenge of digital transformation everywhere.”

- Interview 12

Structural Capacity Imbalances and Empty-Running Rationality

Geographic imbalances cause structural empty running

“This is under-appreciated when people talk about empty running in the public discourse. The stats you see, 20%, 30% empty running, they include a lot of transport types where an empty return leg is literally the product. Gravel trucks, tippers, tankers. Those are structurally empty-return businesses and no platform is going to change that. [...] Where empty running is actually a problem is in the middle, part-loads where theoretically there are backhauls available but matching isn’t happening because of fragmentation, lack of visibility, trust issues. That’s where collaboration and digital platforms could move the needle, if they’re designed right.”

- Interview 12

Customer directives constrain carriers’ ability to planning

“Our own trucking company only have [customer name] goods and we cannot drive anything else. But the subcontractors they are allowed to load for other competitors or so on. We don’t care about that. So, then we just pay them for the volume we use and then they have to find other things to load too.”

- Interview 11

Seasonal volume fluctuations as a barrier to collaboration

“There are moments of slack, of course, seasonality, a cancelled shipment, an empty return leg where there’s just no volume coming back. Theoretically you could backfill

with another customer's goods, and we do think about that. But in practice it's more complex than it sounds."
- Interview 12